

Canonical W configurations - cluster P5

species: electron
polarization cluster: P5
source cluster configurations: 76
canonical configurations collected: 15

canonical display criterion: exactly 3 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

cluster retained-component count distribution:

- 3 components: 15 configurations
- 4 components: 26 configurations
- 5 components: 2 configurations
- 6 components: 31 configurations
- 7 components: 2 configurations

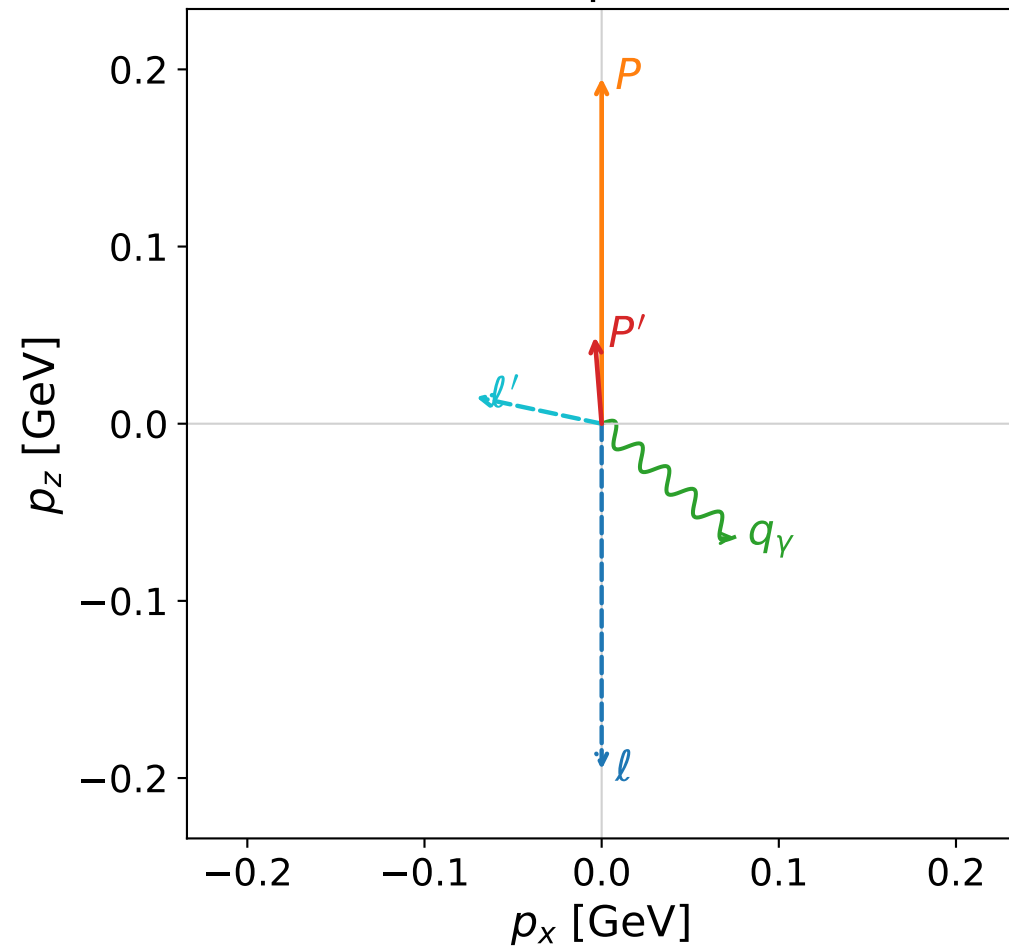
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\W\dw\polarization_cluster_05\combined\canonical_dw_polarization_cluster_05_configurations_electron.csv

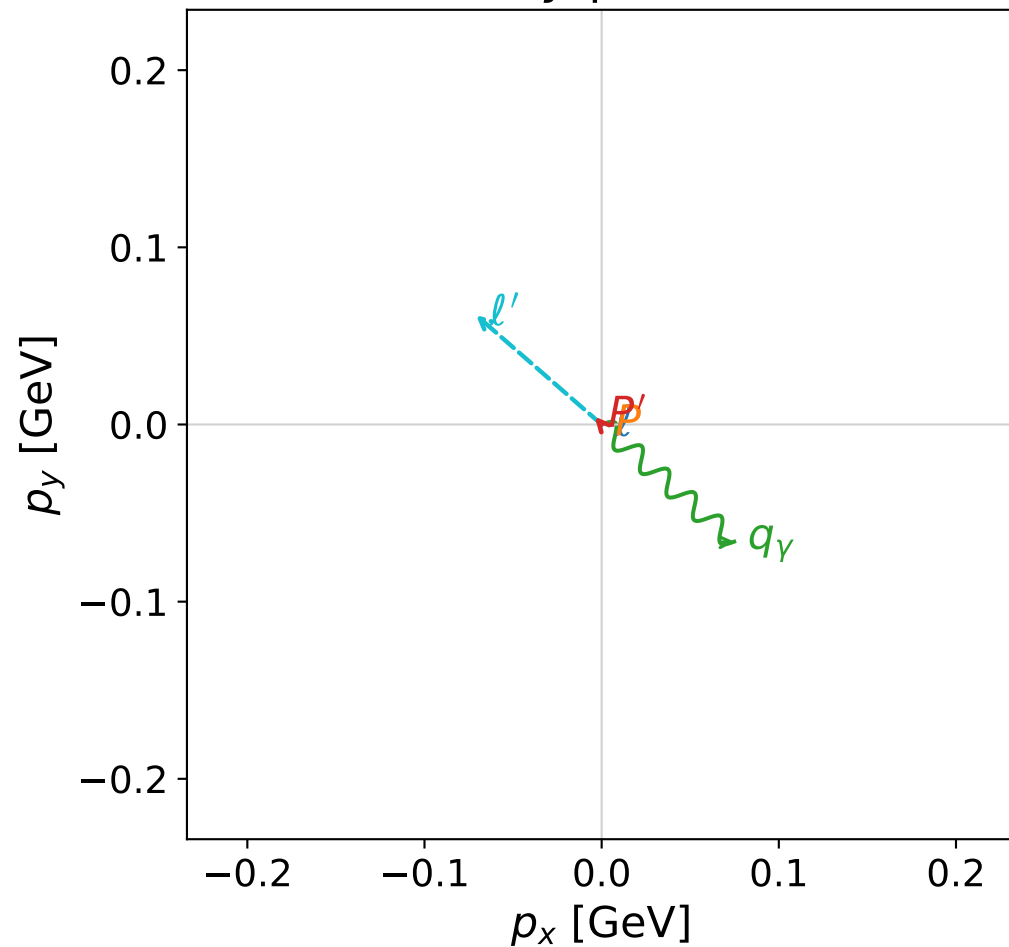
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_05_local_minimum_113

$D_W=0.00028775$

region: polarization_cluster_05_local_minimum_113

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0113

lepton mixing angle: $\alpha_e=2.3605141$ rad

incoming lepton: $-0.710155|+> +0.704046|->$

proton mixing angle: $\alpha_p=3.009653$ rad

incoming proton: $-0.991309|+> +0.131557|->$

$s=1.32973$, $\sqrt{s}=1.15314$, $|\vec{P}|=0.195069$, $|\vec{P}'|=0.0493626$

$\theta_{p'}=0.118924$, $\theta_\gamma=2.1413$, $\phi_{p'}=2.29736$, $\phi_\gamma=5.55919$

$E_\gamma=0.11862$, $\theta_{l'}=1.4121$, $\phi_{l'}=2.42506$

$Q^2=0.0430194$, $x_B=0.157407$, $t=-0.0210141$, $W^2=1.11012$, $y=0.607488$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.1951$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1951$

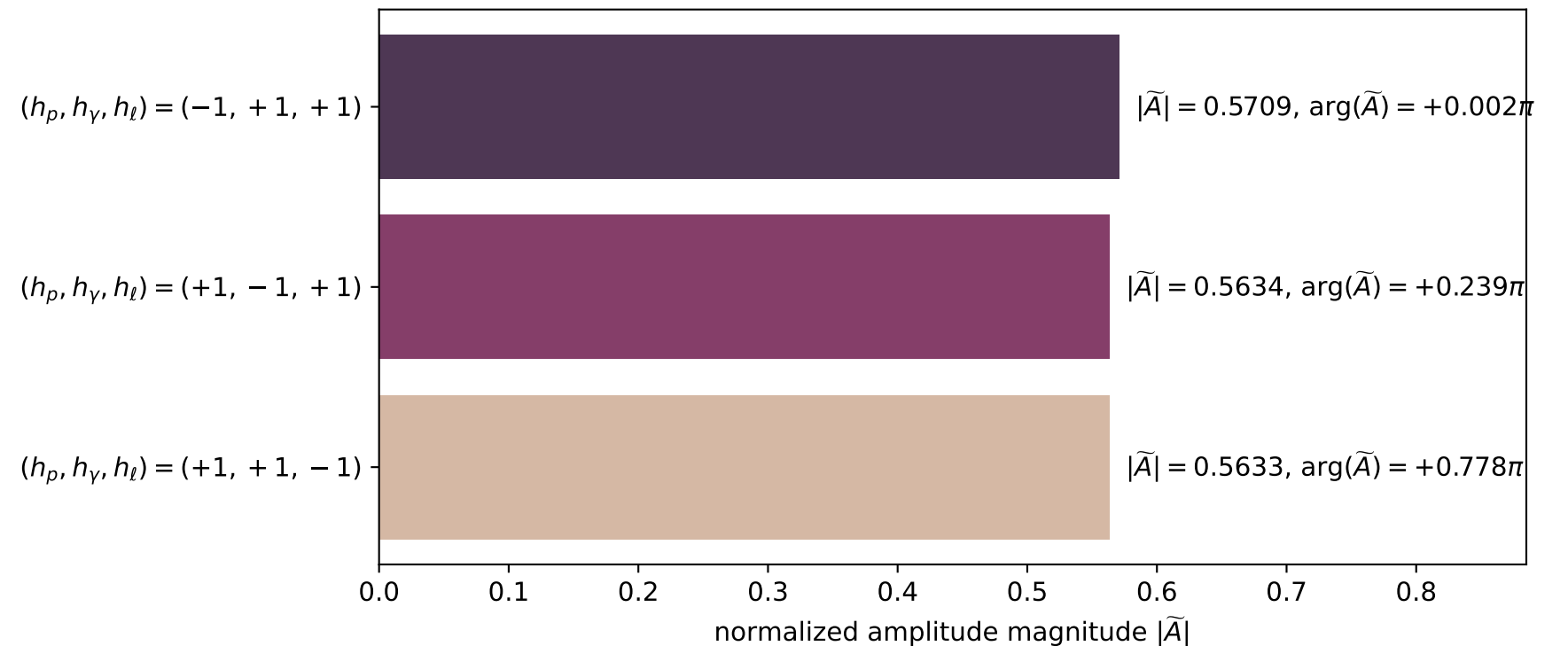
P $E= 0.9581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1951$

l' $E= 0.0952$ $p_x= -0.0709$ $p_y= 0.0618$ $p_z= 0.0150$

P' $E= 0.9393$ $p_x= -0.0039$ $p_y= 0.0044$ $p_z= 0.0490$

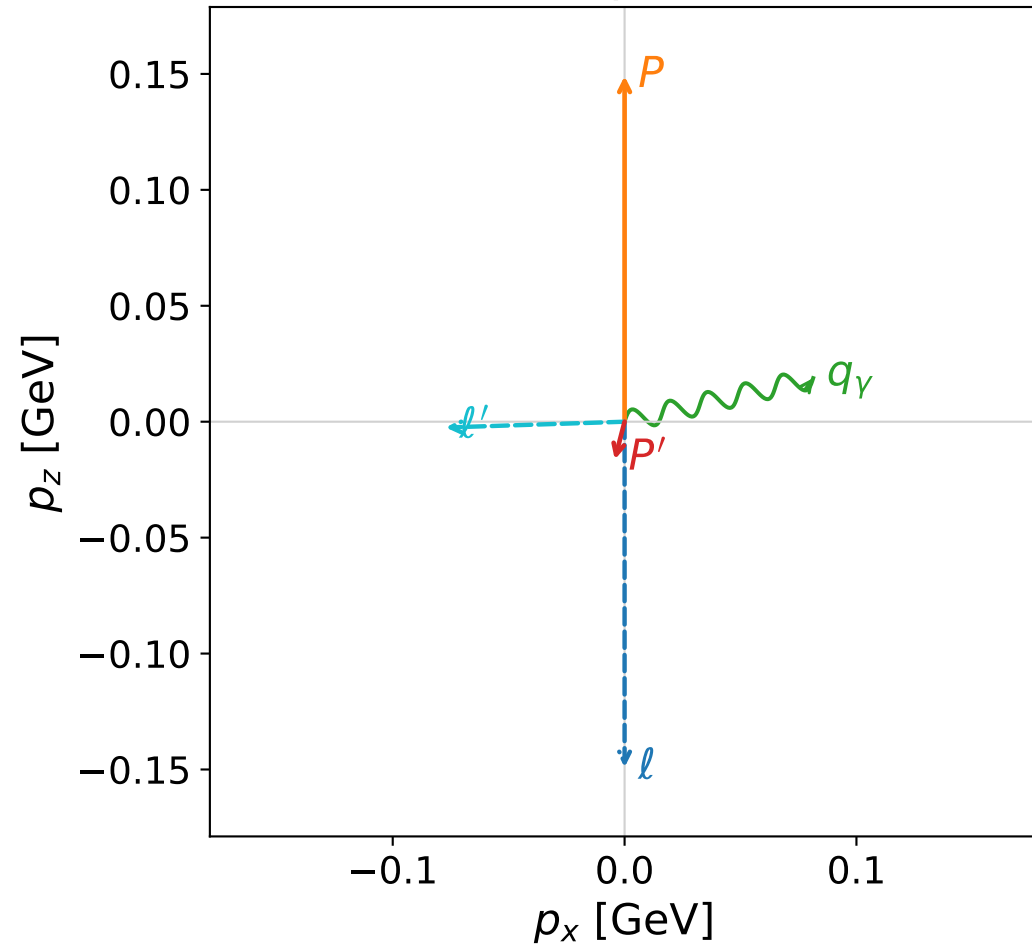
q_γ $E= 0.1186$ $p_x= 0.0748$ $p_y= -0.0661$ $p_z= -0.0641$

Leading normalized final-state amplitudes
(N=3, retained=96.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

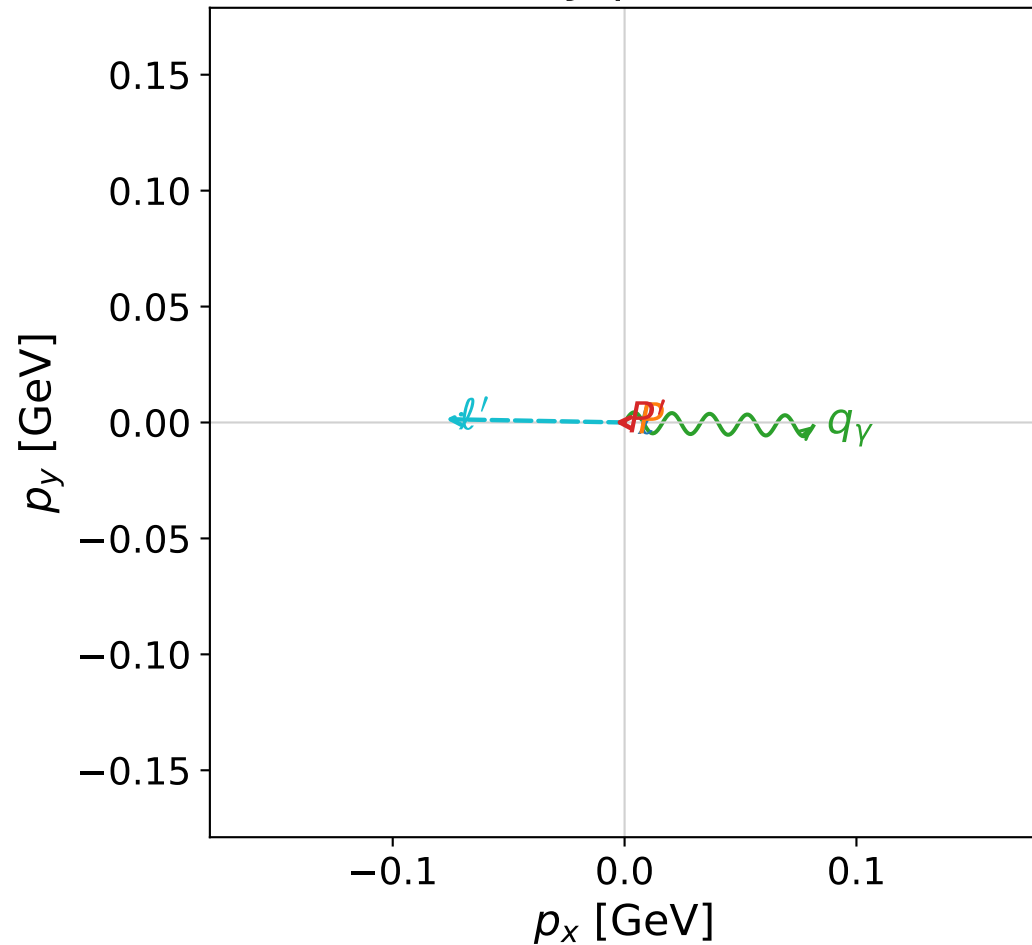


dw_mixing_angles_polarization_05_local_minimum_559
 $D_W=0.00332951$
 region: polarization_cluster_05_local_minimum_559
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0559
 lepton mixing angle: $\alpha_e=2.3556829$ rad
 incoming lepton: $-0.706745|+\rangle +0.707468|-\rangle$
 proton mixing angle: $\alpha_p=3.0233463$ rad
 incoming proton: $-0.993017|+\rangle +0.117971|-\rangle$

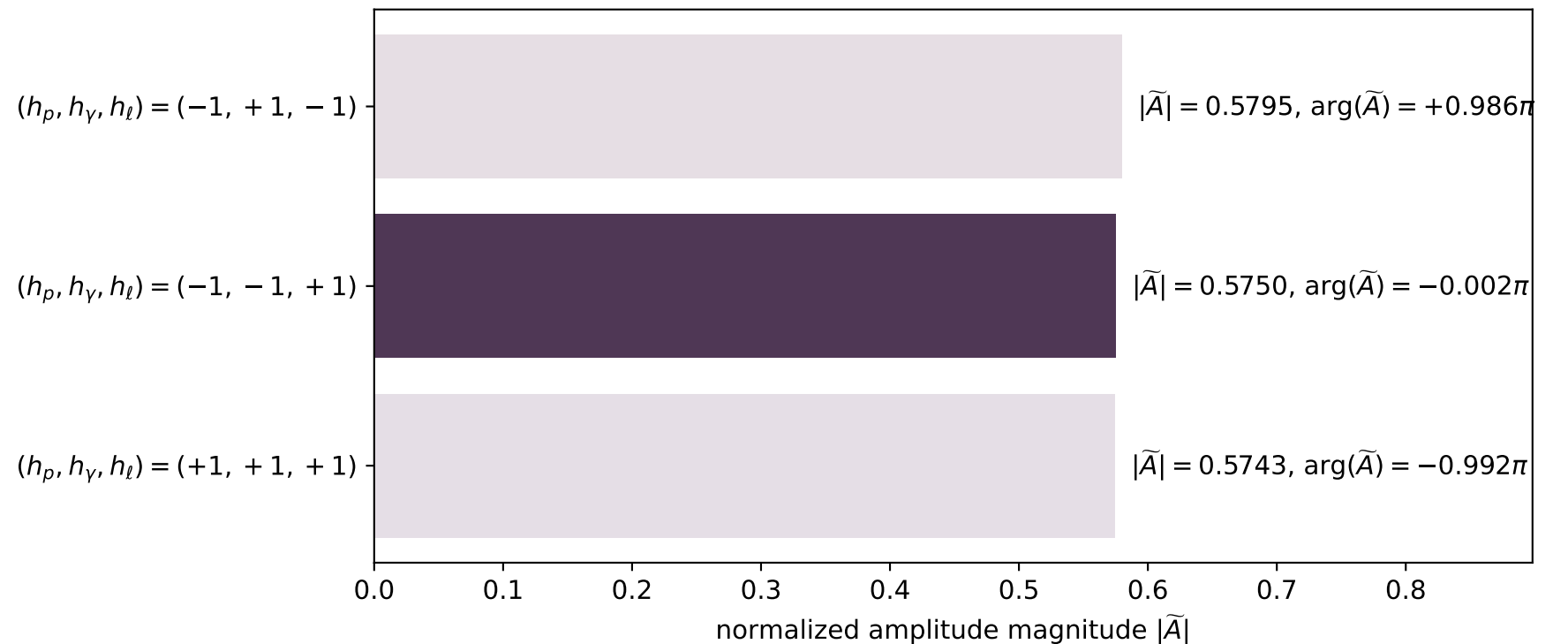
$s=1.20742$, $\sqrt{s}=1.09883$, $|\vec{P}|=0.149056$, $|\vec{P}'|=0.01683$
 $\theta_{p'}=2.88847$, $\theta_\gamma=1.34375$, $\phi_{p'}=3.11512$, $\phi_\gamma=6.26422$
 $E_\gamma=0.083495$, $\theta_{l'}=1.60321$, $\phi_{l'}=3.12304$
 $Q^2=0.0222623$, $x_B=0.123527$, $t=-0.0272232$, $W^2=1.0378$, $y=0.550173$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1491$
 P $E= 0.9498$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1491$
 l' $E= 0.0772$ $p_x= -0.0771$ $p_y= 0.0014$ $p_z= -0.0025$
 P' $E= 0.9382$ $p_x= -0.0042$ $p_y= 0.0001$ $p_z= -0.0163$
 q_γ $E= 0.0835$ $p_x= 0.0813$ $p_y= -0.0015$ $p_z= 0.0188$

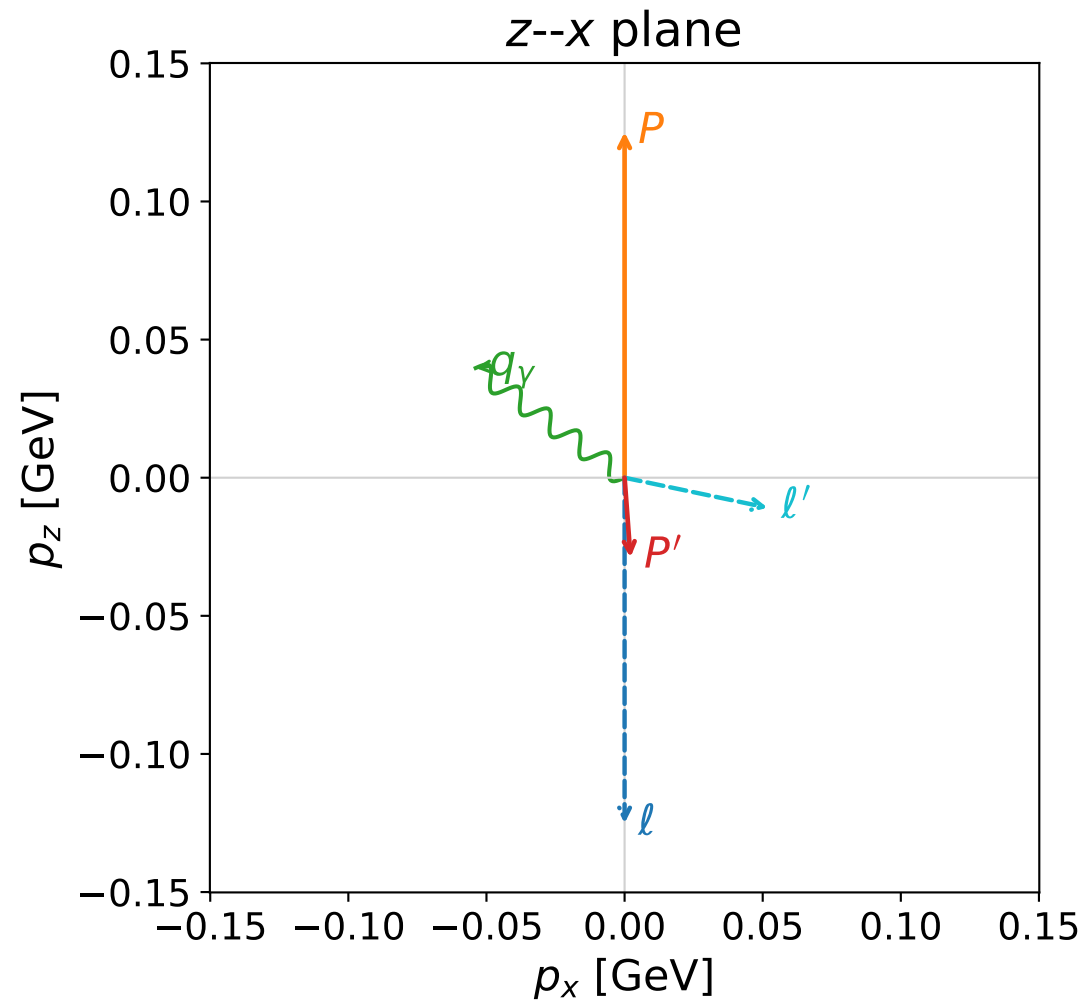
x--y plane



Leading normalized final-state amplitudes
 (N=3, retained=99.6%)



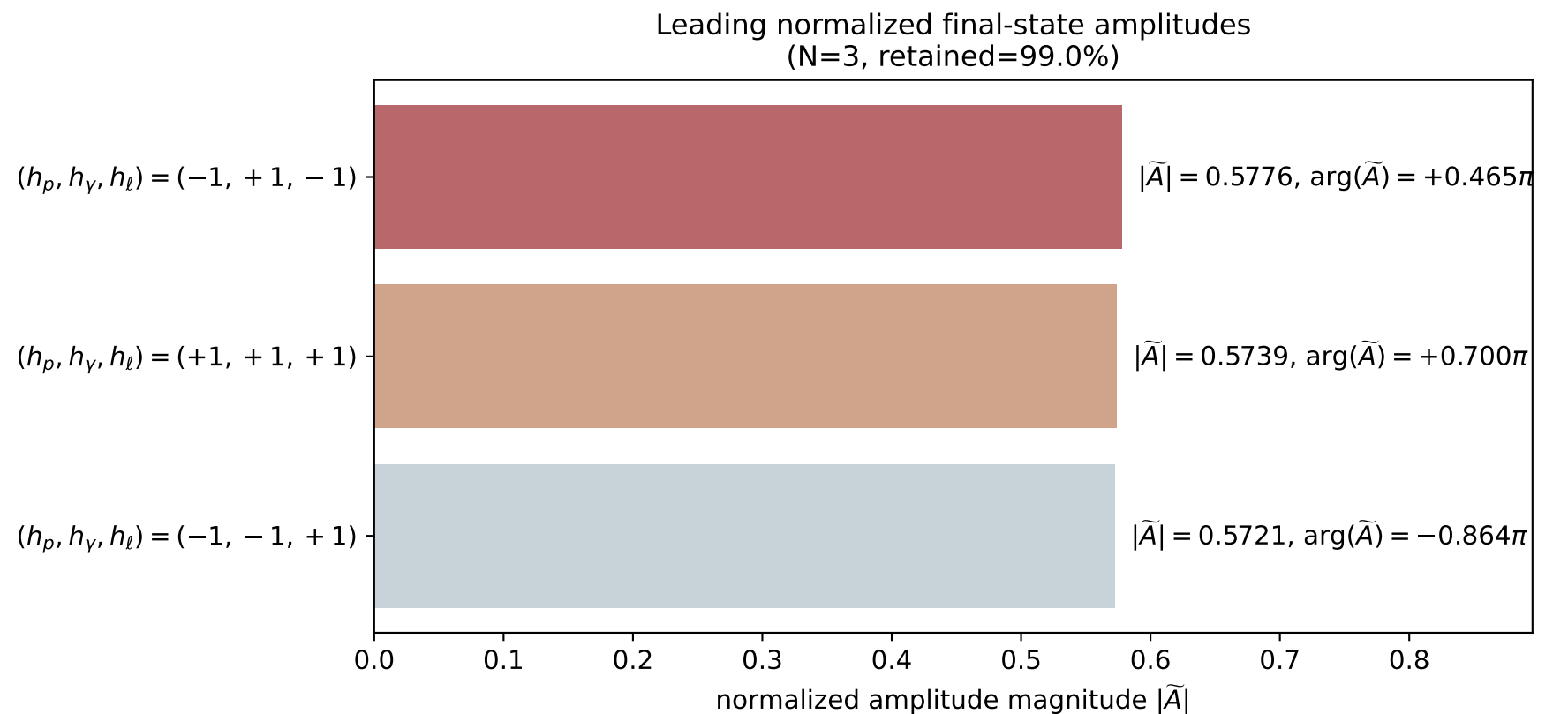
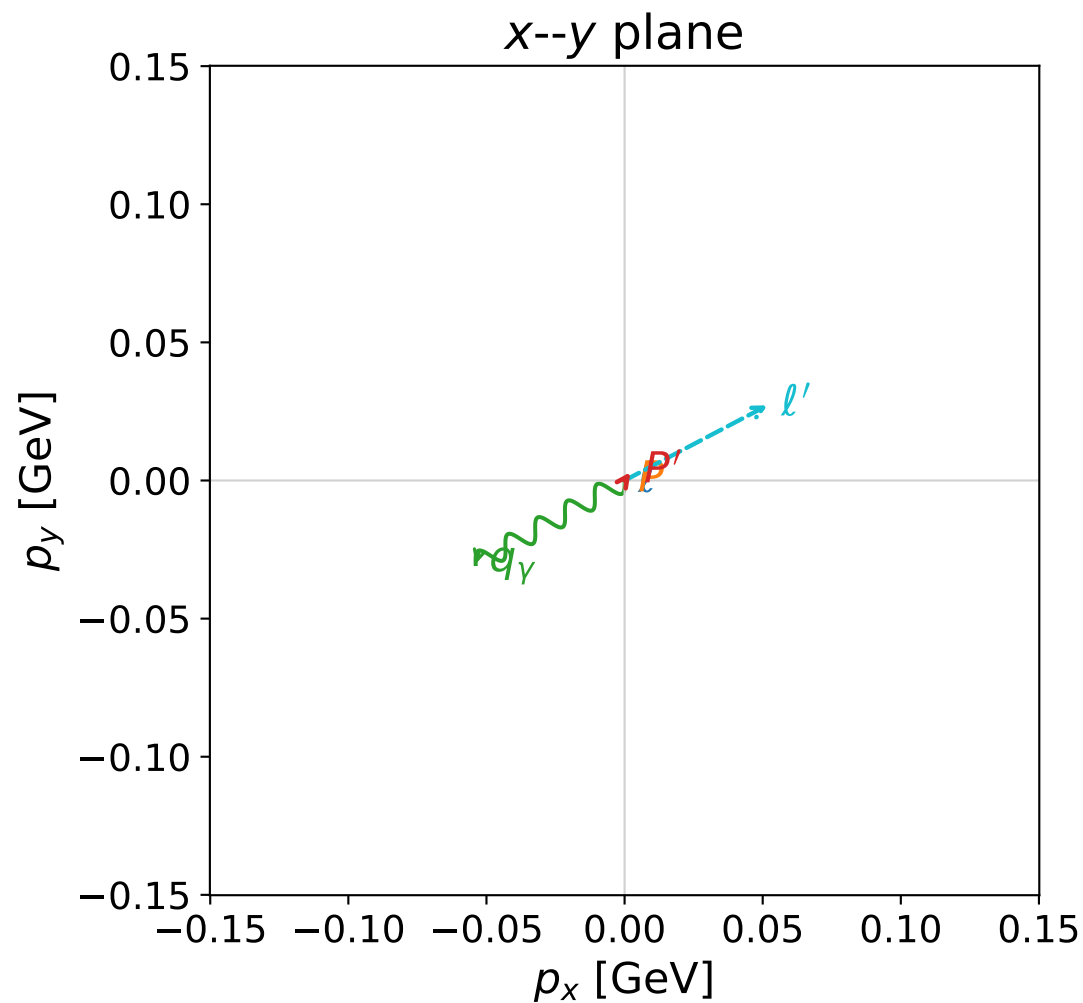
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



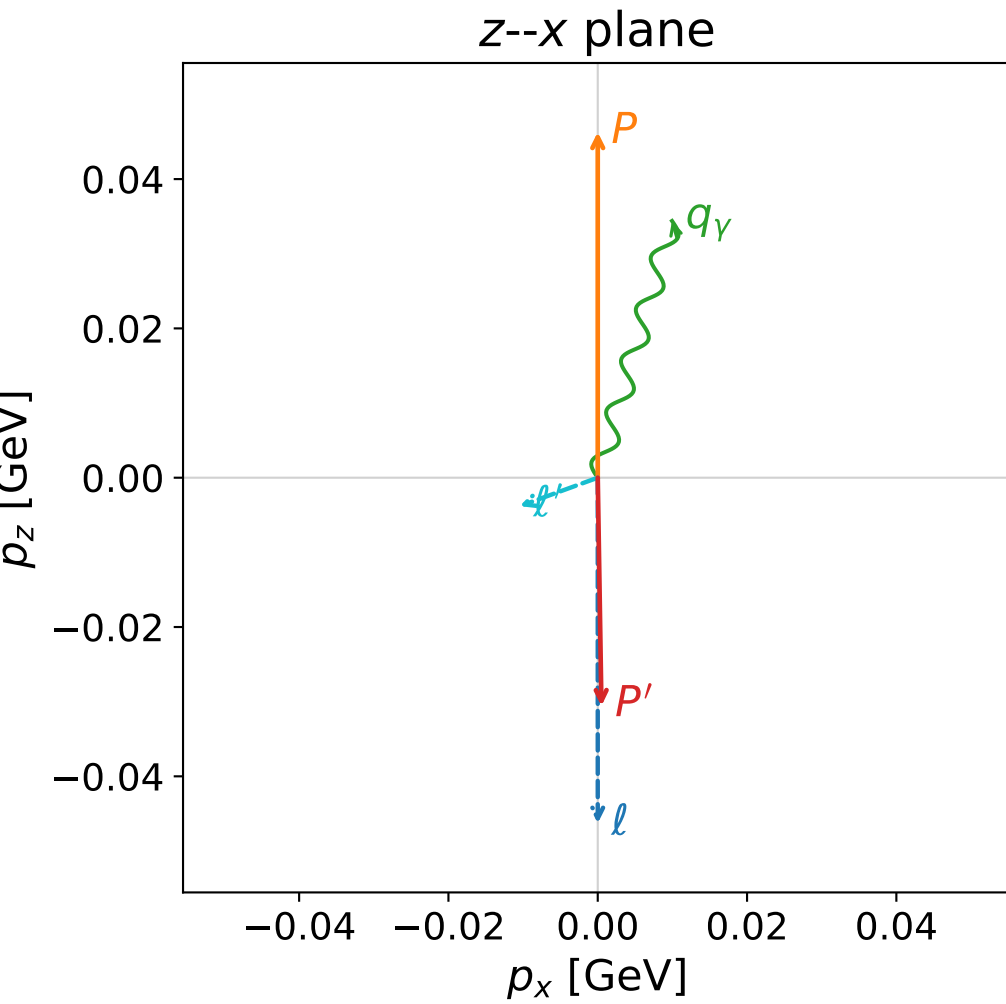
```
dw_mixing_angles_polarization_05_local_minimum_569
D_W=0.00349548
region: polarization_cluster_05_local_minimum_569
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0569
lepton mixing angle: alpha_e=2.3431432 rad
incoming lepton: -0.697818|+> +0.716275|->
proton mixing angle: alpha_p=0.13845223 rad
incoming proton: 0.990431|+> +0.13801|->
```

```
s=1.14793, sqrt(s)=1.07141, |P|=0.125106, |P'|=0.0290871
theta_p=3.01447, theta_gamma=0.99976, phi_p=0.954904, phi_gamma=3.65282
E_gamma=0.0734631, theta_l=1.75427, phi_l=0.484158
Q^2=0.012171, x_B=0.0796754, t=-0.023655, W^2=1.02043, y=0.569815
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  0.1251 p_x=  0.0000 p_y=  0.0000 p_z= -0.1251
P      E=  0.9463 p_x=  0.0000 p_y=  0.0000 p_z=  0.1251
l'     E=  0.0595 p_x=  0.0518 p_y=  0.0272 p_z= -0.0109
P'     E=  0.9385 p_x=  0.0021 p_y=  0.0030 p_z= -0.0289
q_gamma E=  0.0735 p_x= -0.0539 p_y= -0.0302 p_z=  0.0397
```



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



dw_mixing_angles_polarization_05_local_minimum_606

D_W=0.00424053

region: polarization_cluster_05_local_minimum_606

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0606

lepton mixing angle: $\alpha_e=2.3490452$ rad

incoming lepton: $-0.702033|+> +0.712144|->$

proton mixing angle: $\alpha_p=3.0187472$ rad

incoming proton: $-0.992464|+> +0.122537|->$

$s=0.971085$, $\sqrt{s}=0.985437$, $|\vec{P}|=0.0462921$, $|\vec{P}'|=0.0305854$

$\theta_{p'}=3.12485$, $\theta_\gamma=0.281828$, $\phi_{p'}=0.254548$, $\phi_\gamma=6.24302$

$E_\gamma=0.0358045$, $\theta_{\ell'}=1.92053$, $\phi_{\ell'}=3.11566$

$Q^2=0.000677509$, $x_B=0.009682$, $t=-0.00590934$, $W^2=0.949143$, $y=0.766938$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

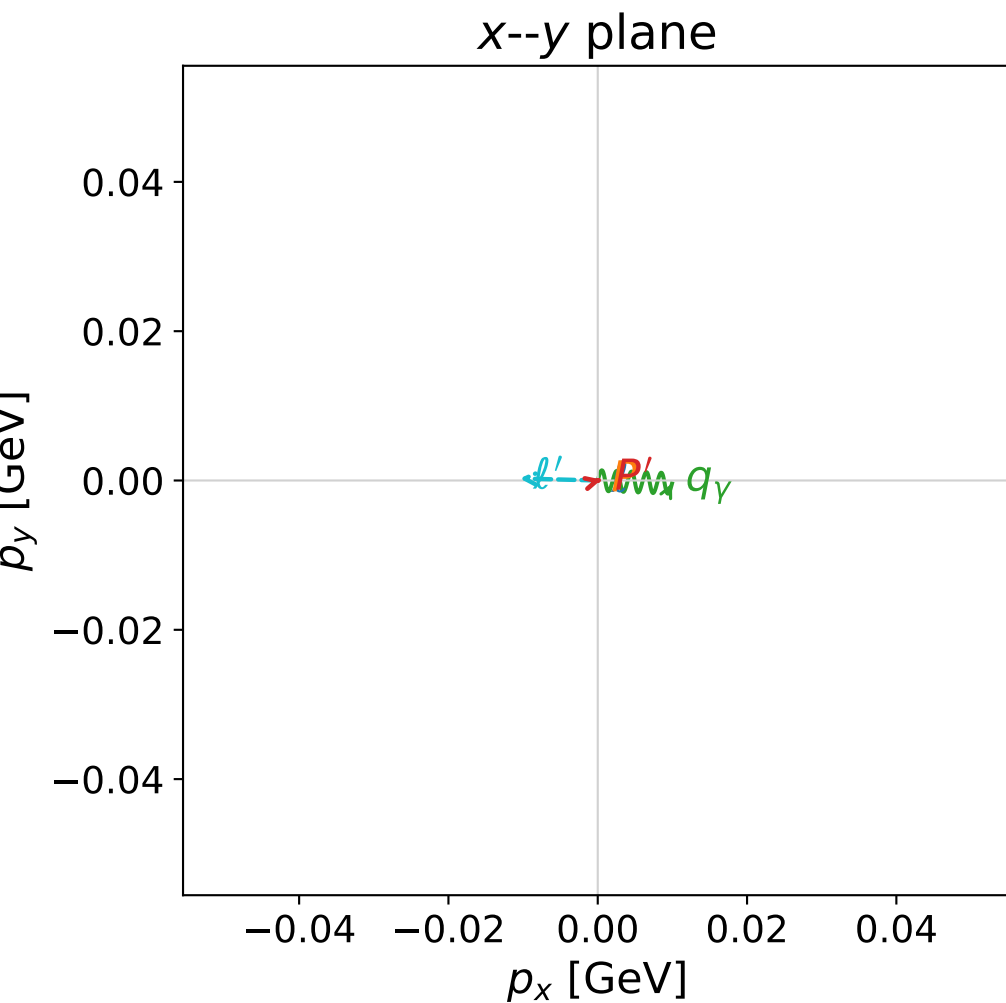
ℓ $E= 0.0463$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0463$

P $E= 0.9391$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0463$

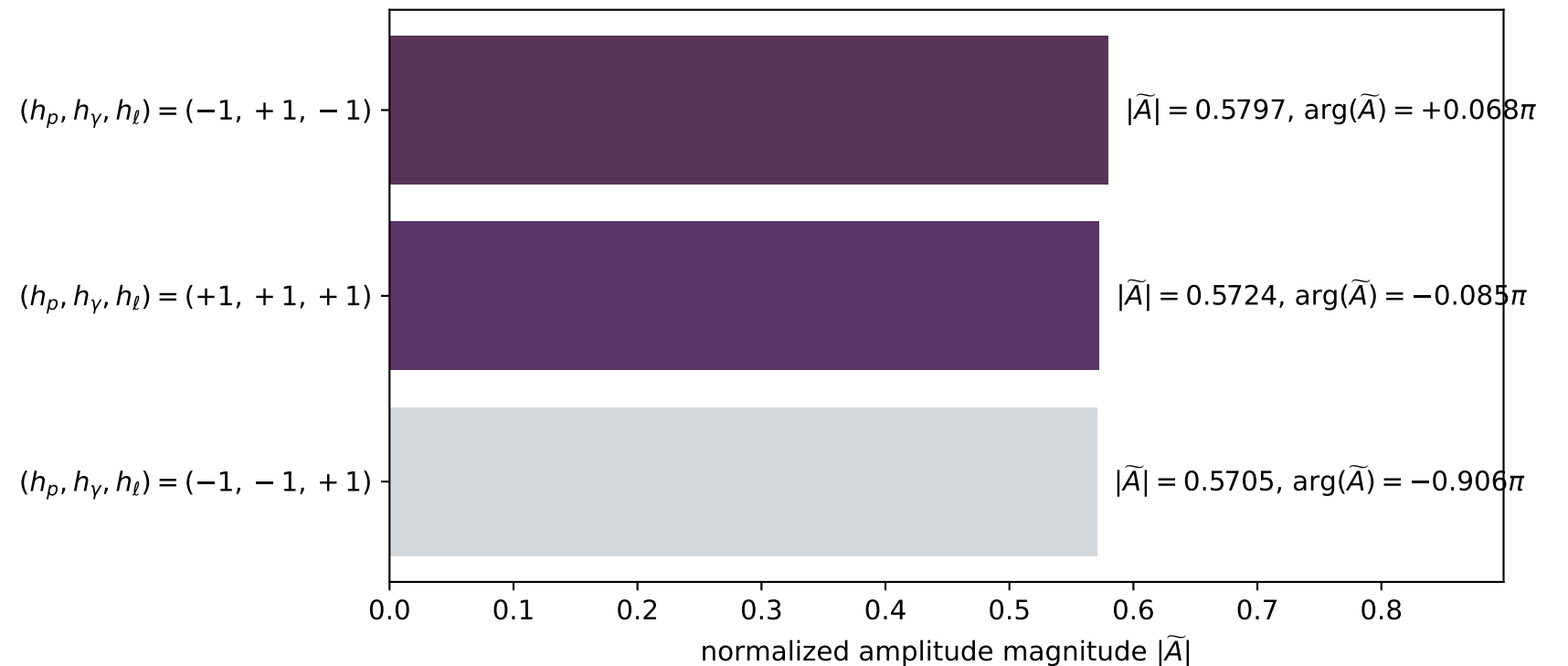
ℓ' $E= 0.0111$ $p_x= -0.0104$ $p_y= 0.0003$ $p_z= -0.0038$

P' $E= 0.9385$ $p_x= 0.0005$ $p_y= 0.0001$ $p_z= -0.0306$

q_γ $E= 0.0358$ $p_x= 0.0099$ $p_y= -0.0004$ $p_z= 0.0344$

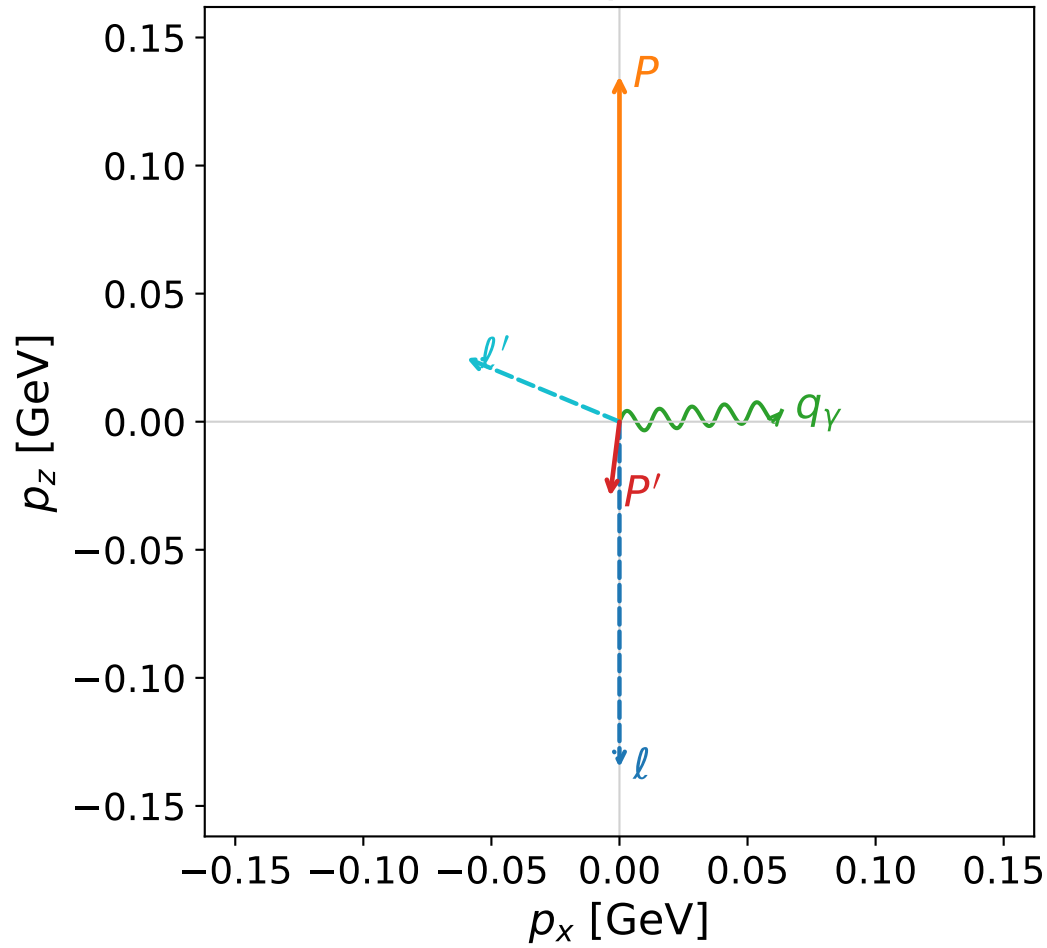


Leading normalized final-state amplitudes
(N=3, retained=98.9%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

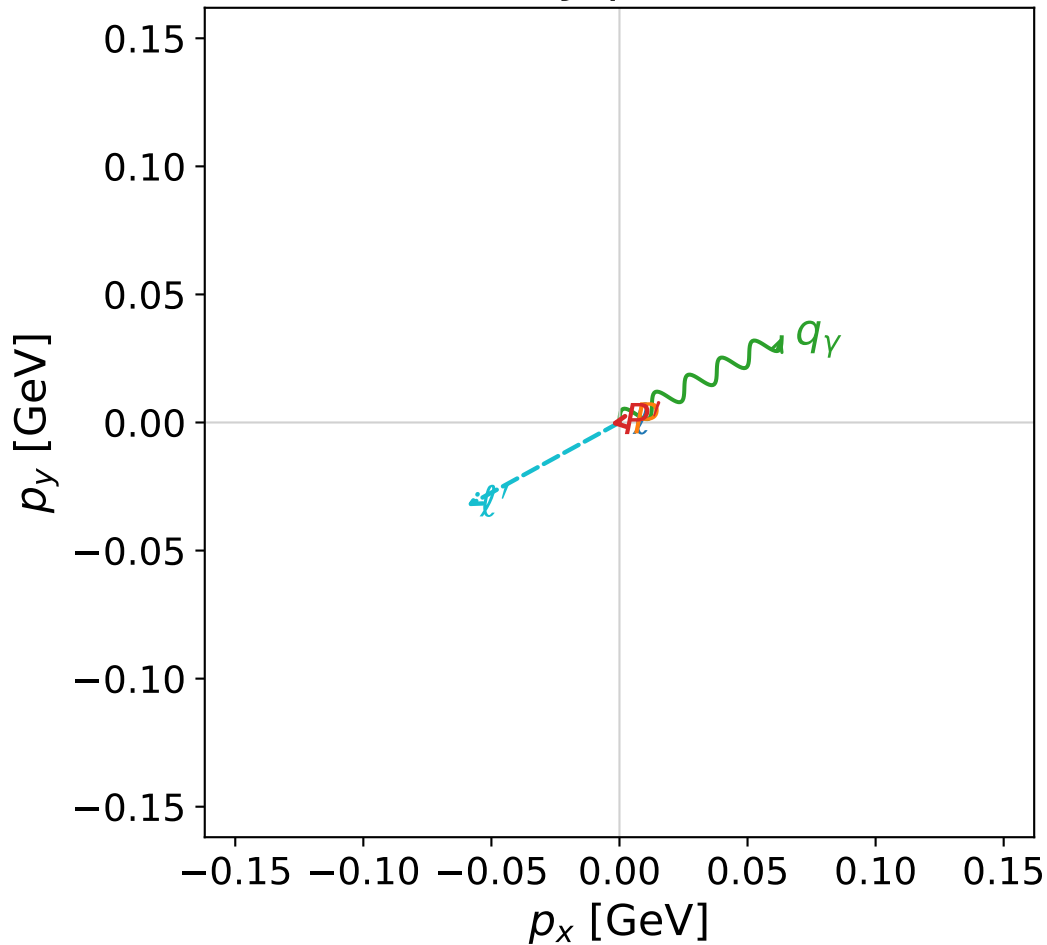


dw_mixing_angles_polarization_05_local_minimum_728
 $D_W=0.00832553$
 region: polarization_cluster_05_local_minimum_728
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0728
 lepton mixing angle: $\alpha_e=2.3656503$ rad
 incoming lepton: $-0.713761|+\rangle +0.700389|-\rangle$
 proton mixing angle: $\alpha_p=3.0654637$ rad
 incoming proton: $-0.997104|+\rangle +0.0760554|-\rangle$

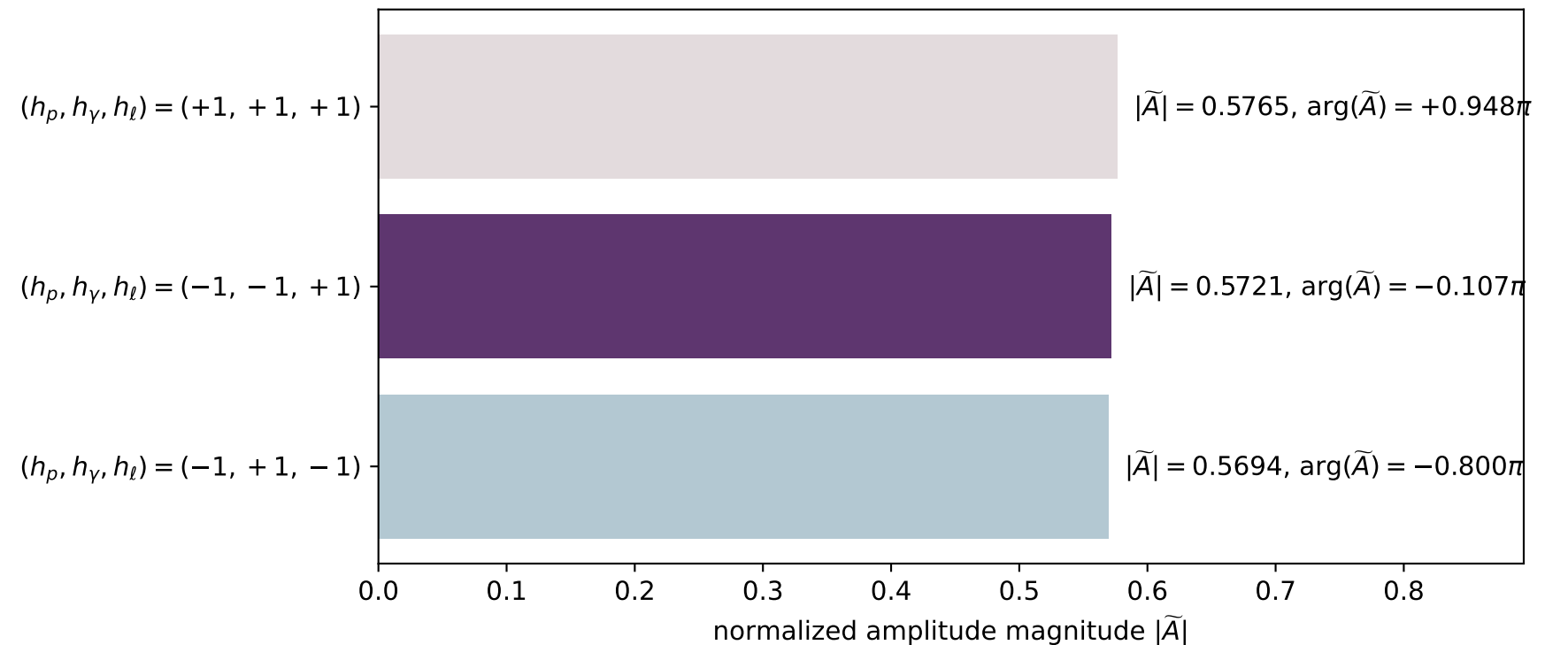
$s=1.17197$, $\sqrt{s}=1.08258$, $|\vec{P}|=0.134923$, $|\vec{P}'|=0.0293$
 $\theta_{p'}=3.01563$, $\theta_\gamma=1.51191$, $\phi_{p'}=3.29583$, $\phi_\gamma=0.483699$
 $E_\gamma=0.0716602$, $\theta_{l'}=1.22072$, $\phi_{l'}=3.64279$
 $Q^2=0.0262585$, $x_B=0.162589$, $t=-0.0268219$, $W^2=1.01509$, $y=0.552843$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1349$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1349$
 P $E= 0.9477$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1349$
 l' $E= 0.0725$ $p_x= -0.0597$ $p_y= -0.0327$ $p_z= 0.0249$
 P' $E= 0.9385$ $p_x= -0.0036$ $p_y= -0.0006$ $p_z= -0.0291$
 q_γ $E= 0.0717$ $p_x= 0.0633$ $p_y= 0.0333$ $p_z= 0.0042$

x--y plane

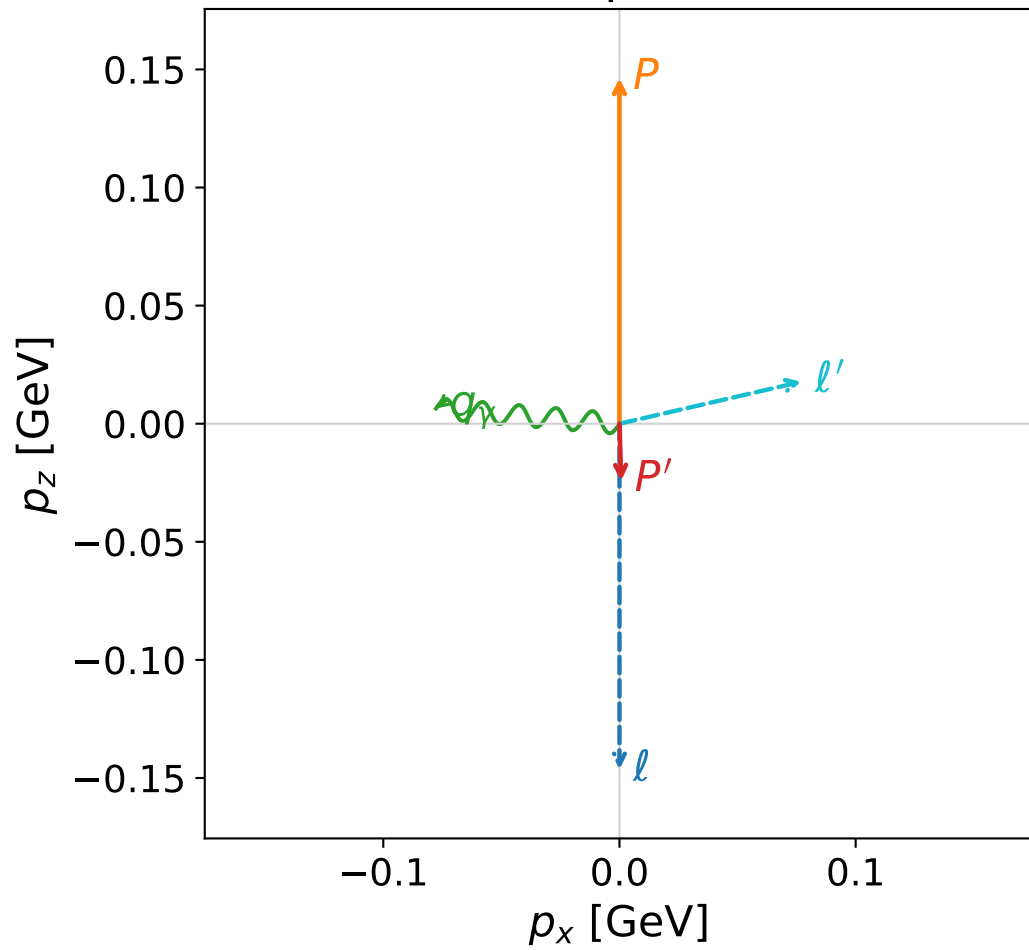


Leading normalized final-state amplitudes
 (N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



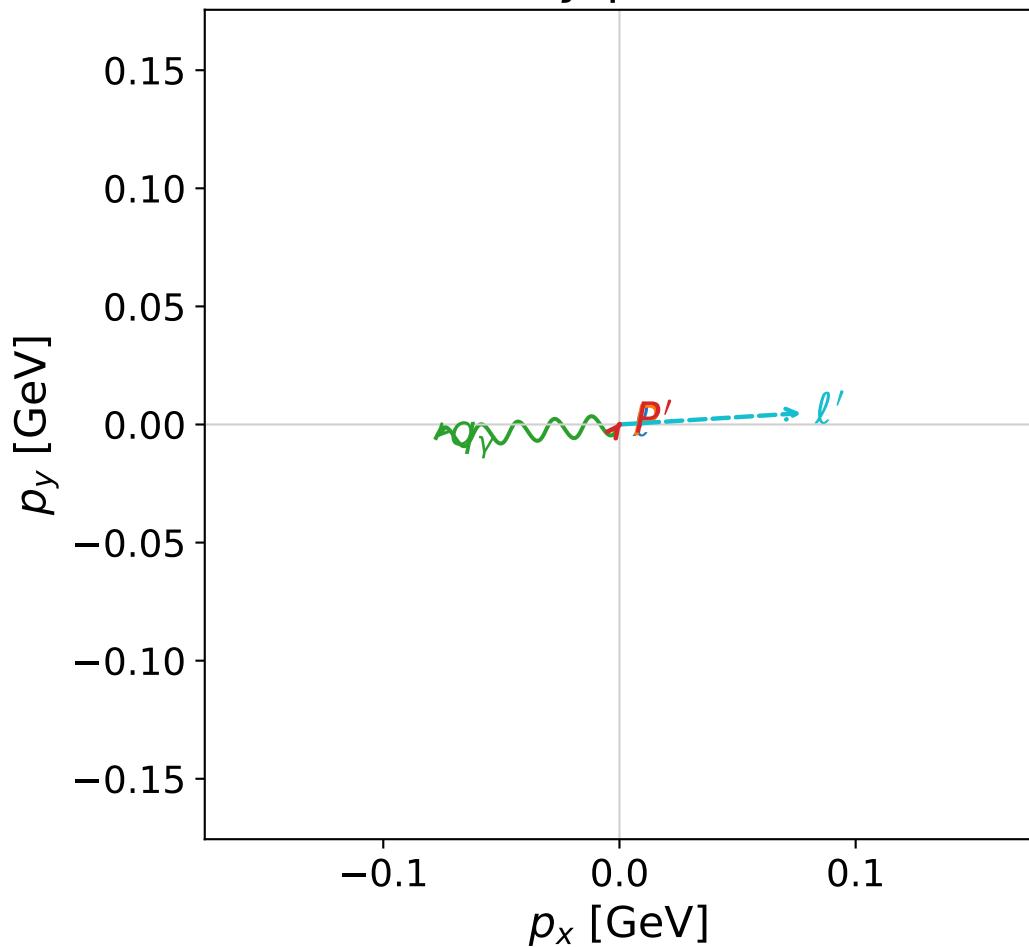
```
dw_mixing_angles_polarization_05_local_minimum_731
D_W=0.00839324
region: polarization_cluster_05_local_minimum_731
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0731
lepton mixing angle: alpha_e=2.3650174 rad
incoming lepton: -0.713318|+> +0.700841|->
proton mixing angle: alpha_p=0.18228078 rad
incoming proton: 0.983433|+> +0.181273|->
```

$s=1.20048$, $\sqrt{s}=1.09567$, $|\vec{P}|=0.146321$, $|\vec{P}'|=0.024235$
 $\theta_{p'}=3.0953$, $\theta_\gamma=1.48825$, $\phi_{p'}=0.920045$, $\phi_\gamma=3.21463$
 $E_\gamma=0.0781683$, $\theta_{\ell'}=1.34453$, $\phi_{\ell'}=0.0621503$
 $Q^2=0.0283707$, $x_B=0.161665$, $t=-0.0289599$, $W^2=1.02696$, $y=0.547318$

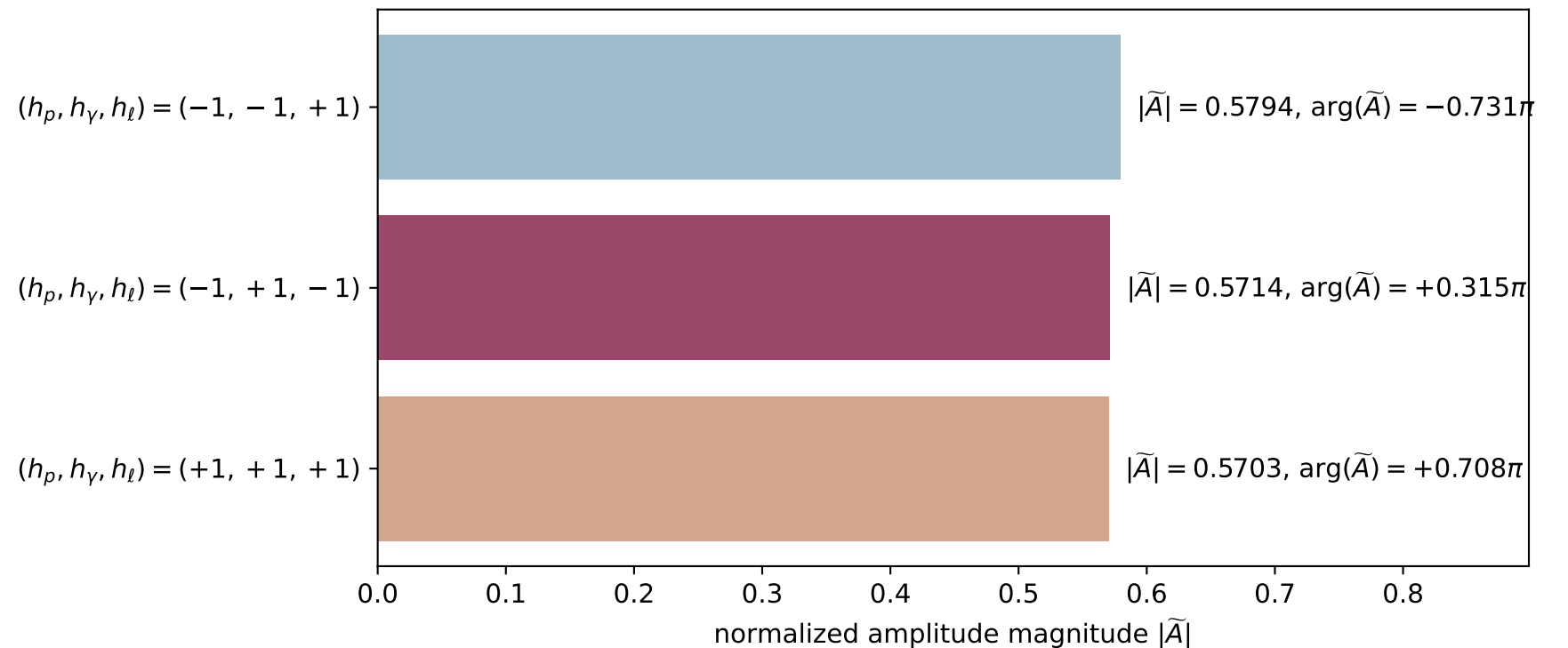
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1463	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1463
P	$E=$	0.9493	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1463
ℓ'	$E=$	0.0792	$p_x=$	0.0770	$p_y=$	0.0048	$p_z=$	0.0178
P'	$E=$	0.9383	$p_x=$	0.0007	$p_y=$	0.0009	$p_z=$	-0.0242
q_γ	$E=$	0.0782	$p_x=$	-0.0777	$p_y=$	-0.0057	$p_z=$	0.0064

x--y plane

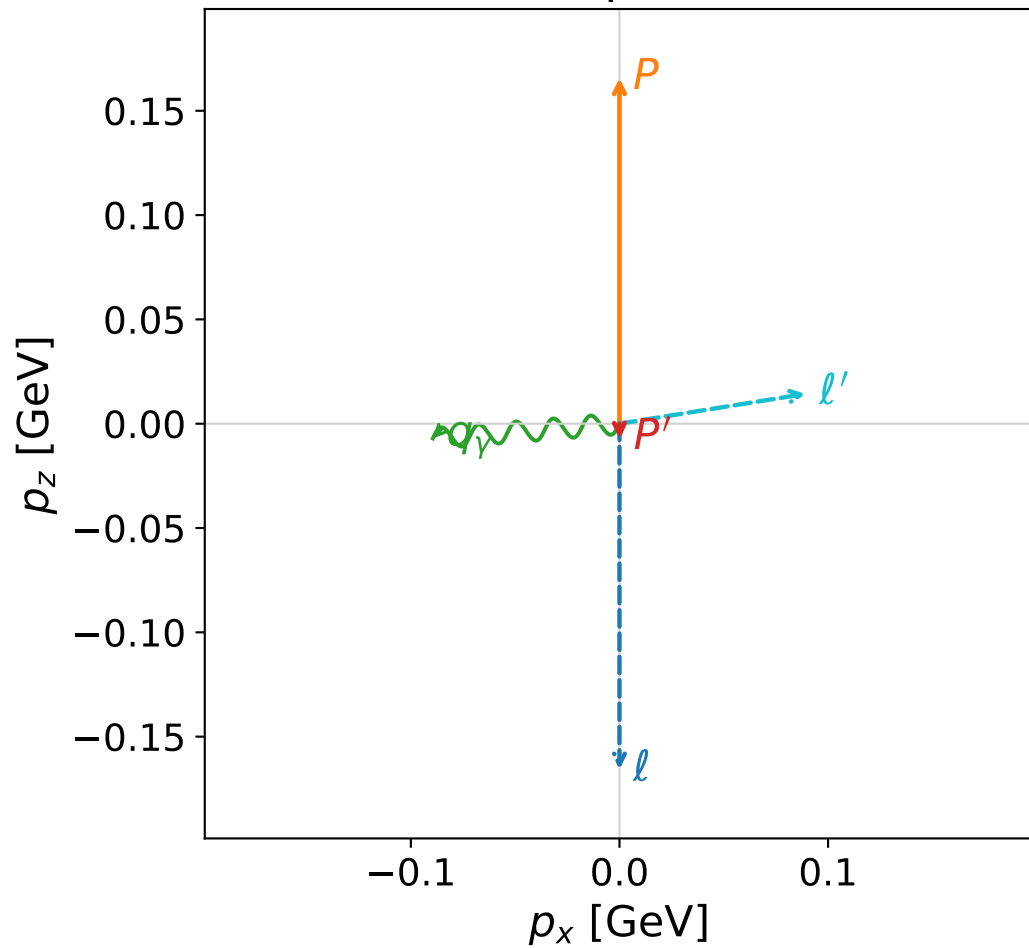


Leading normalized final-state amplitudes
(N=3, retained=98.8%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_05_local_minimum_748

$D_W=0.00890618$

region: polarization_cluster_05_local_minimum_748

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0748

lepton mixing angle: $\alpha_e=2.358533$ rad

incoming lepton: $-0.708758|+\rangle +0.705451|-\rangle$

proton mixing angle: $\alpha_p=0.19379091$ rad

incoming proton: $0.981281|+\rangle +0.19258|-\rangle$

$s=1.25031$, $\sqrt{s}=1.11817$, $|\vec{P}|=0.165657$, $|\vec{P}'|=0.00726959$

$\theta_{p'}=3.06736$, $\theta_\gamma=1.64926$, $\phi_{p'}=1.07968$, $\phi_\gamma=3.17433$

$E_\gamma=0.0897761$, $\theta_{\ell'}=1.41203$, $\phi_{\ell'}=0.0275043$

$Q^2=0.0346737$, $x_B=0.170769$, $t=-0.0296871$, $W^2=1.04822$, $y=0.548077$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1657$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1657$

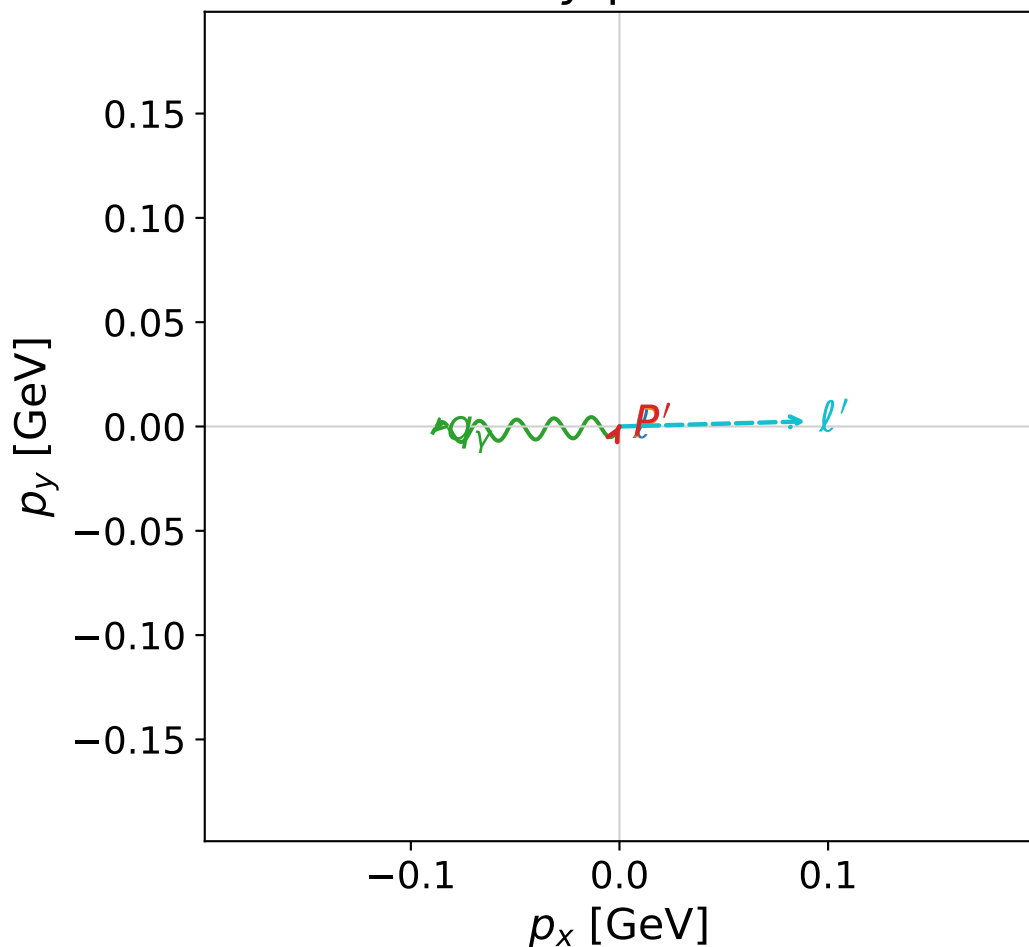
P $E= 0.9525$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1657$

ℓ' $E= 0.0904$ $p_x= 0.0892$ $p_y= 0.0025$ $p_z= 0.0143$

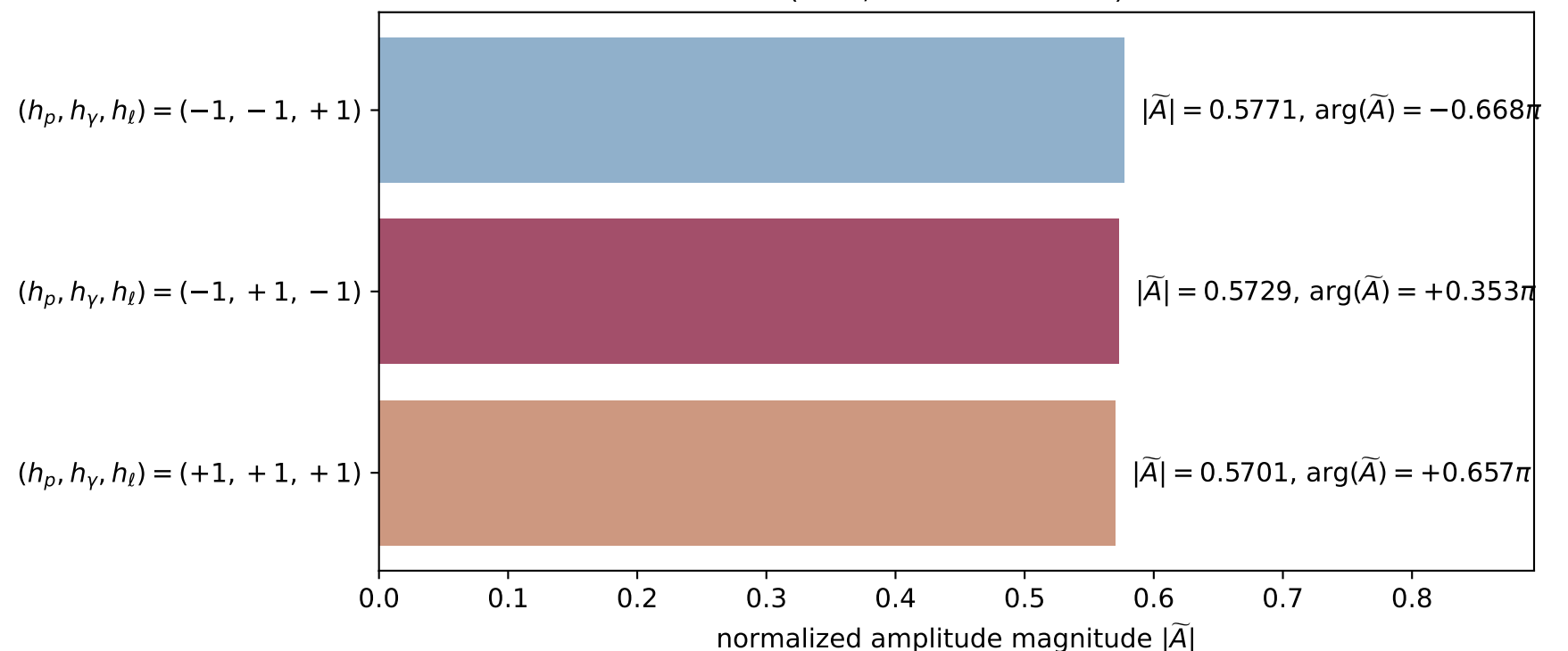
P' $E= 0.9380$ $p_x= 0.0003$ $p_y= 0.0005$ $p_z= -0.0072$

q_γ $E= 0.0898$ $p_x= -0.0895$ $p_y= -0.0029$ $p_z= -0.0070$

x--y plane

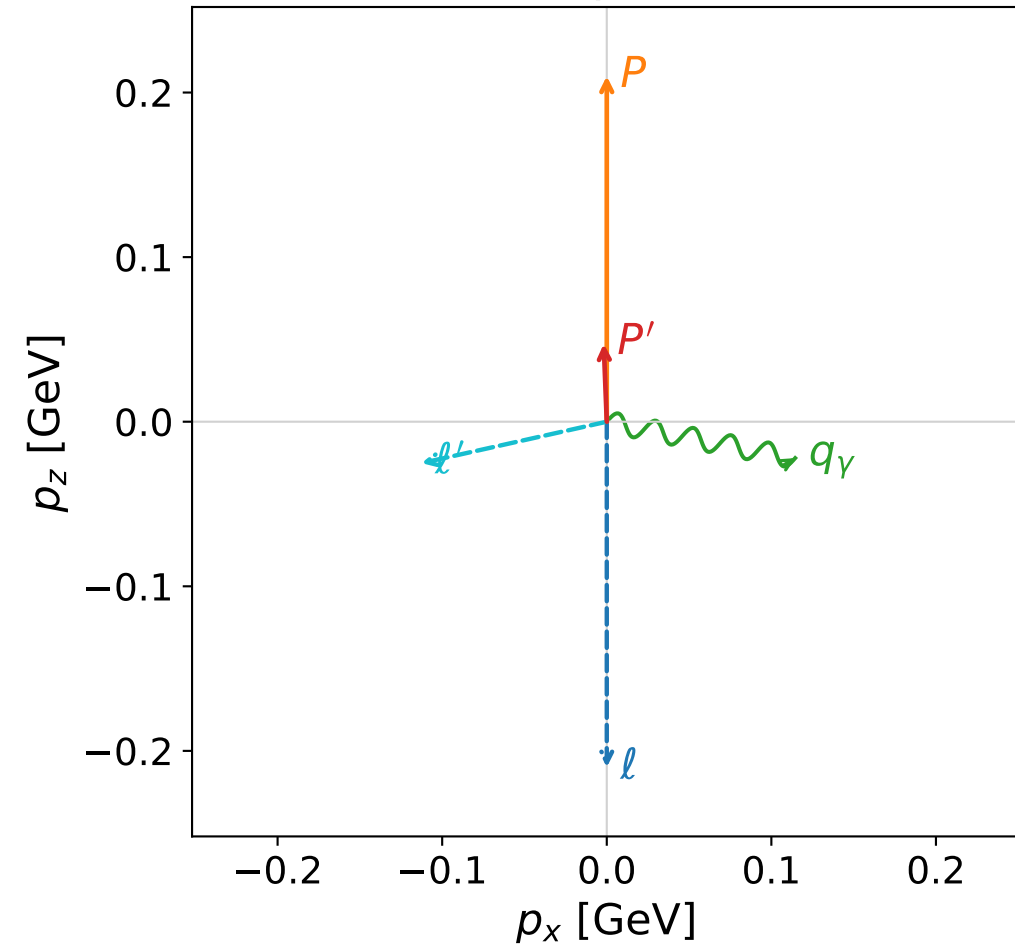


Leading normalized final-state amplitudes
(N=3, retained=98.6%)

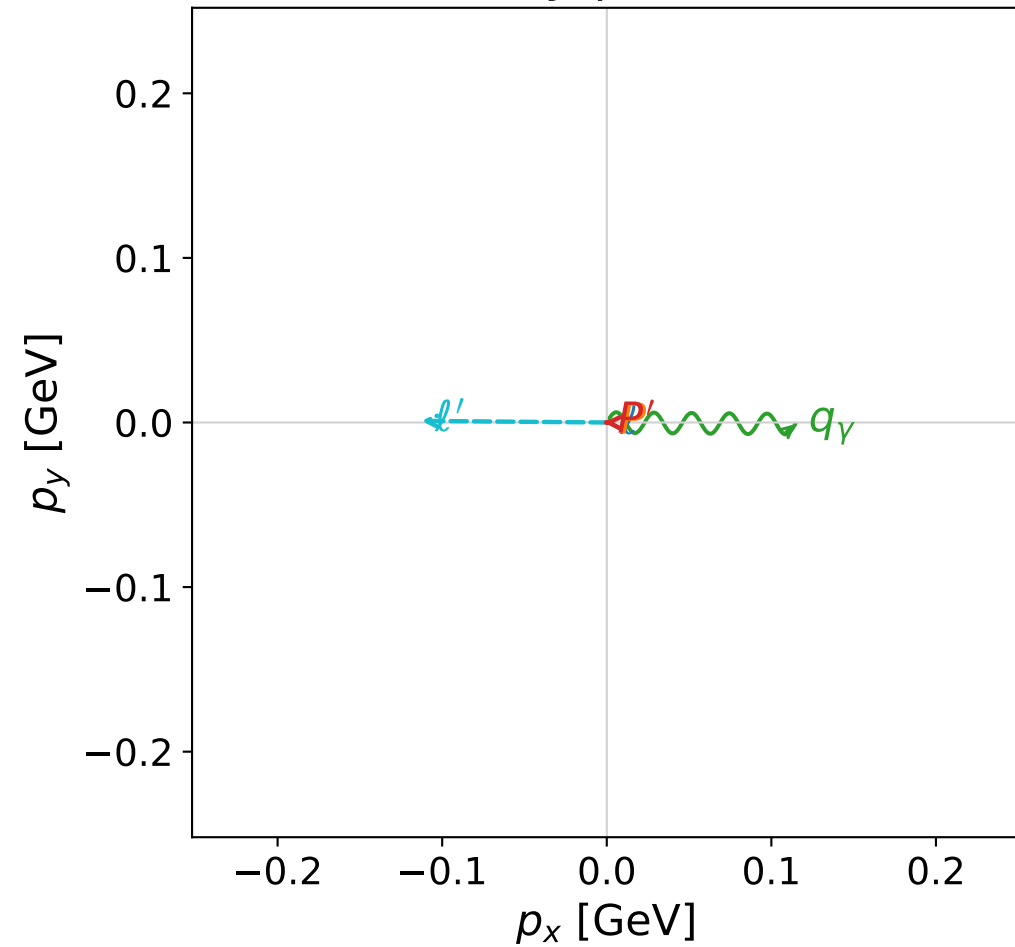


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

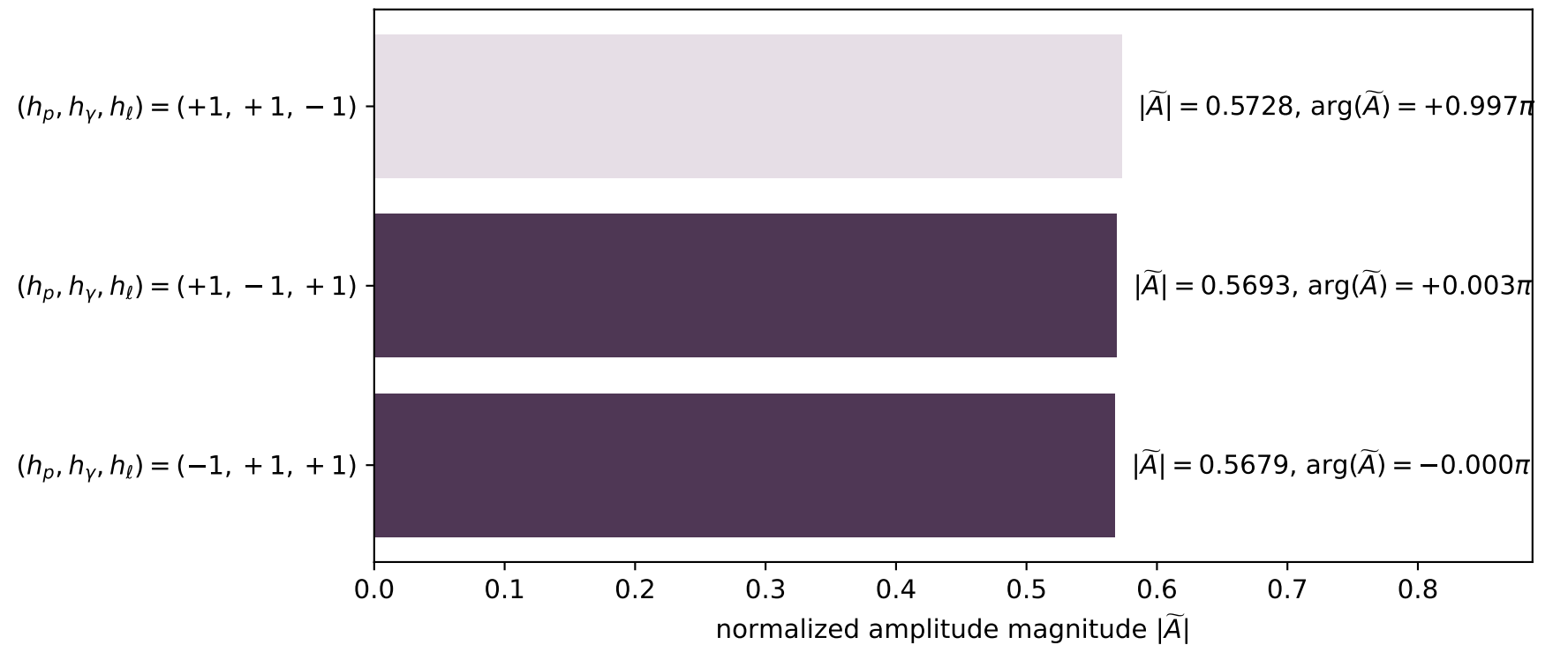


dw_mixing_angles_polarization_05_local_minimum_752
 $D_W=0.00917587$
 region: polarization_cluster_05_local_minimum_752
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0752
 lepton mixing angle: $\alpha_e=2.3432894$ rad
 incoming lepton: $-0.697923|+\rangle + 0.716173|-\rangle$
 proton mixing angle: $\alpha_p=3.0217874$ rad
 incoming proton: $-0.992832|+\rangle + 0.119519|-\rangle$

$s=1.37168$, $\sqrt{s}=1.17119$, $|\vec{P}|=0.209973$, $|\vec{P}'|=0.0473169$
 $\theta_{p'}=0.0392699$, $\theta_\gamma=1.76154$, $\phi_{p'}=3.12521$, $\phi_\gamma=6.27439$
 $E_\gamma=0.116593$, $\theta_{l'}=1.79072$, $\phi_{l'}=3.13292$
 $Q^2=0.0378896$, $x_B=0.14606$, $t=-0.0259872$, $W^2=1.10137$, $y=0.527436$

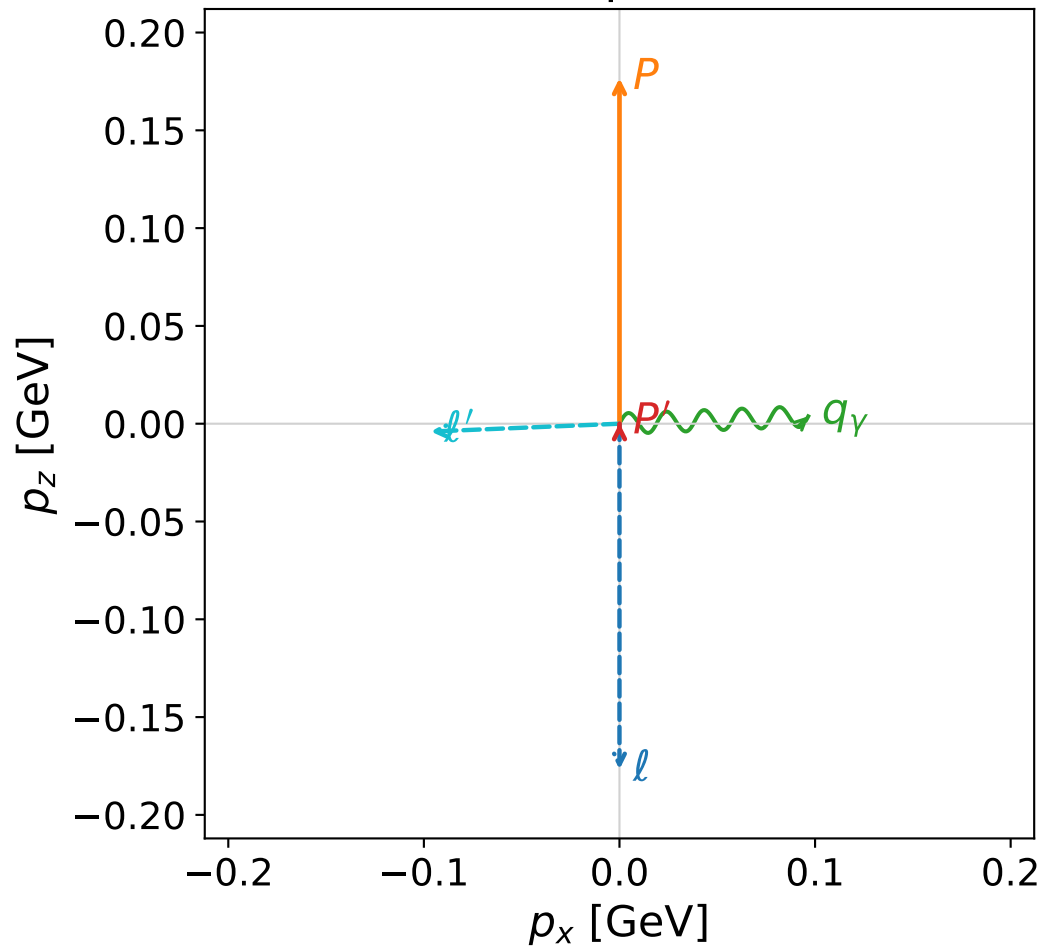
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.2100$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2100$
 P $E=0.9612$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2100$
 l' $E=0.1154$ $p_x=-0.1126$ $p_y=0.0010$ $p_z=-0.0252$
 P' $E=0.9392$ $p_x=-0.0019$ $p_y=0.0000$ $p_z=0.0473$
 q_γ $E=0.1166$ $p_x=0.1145$ $p_y=-0.0010$ $p_z=-0.0221$

Leading normalized final-state amplitudes
 (N=3, retained=97.5%)

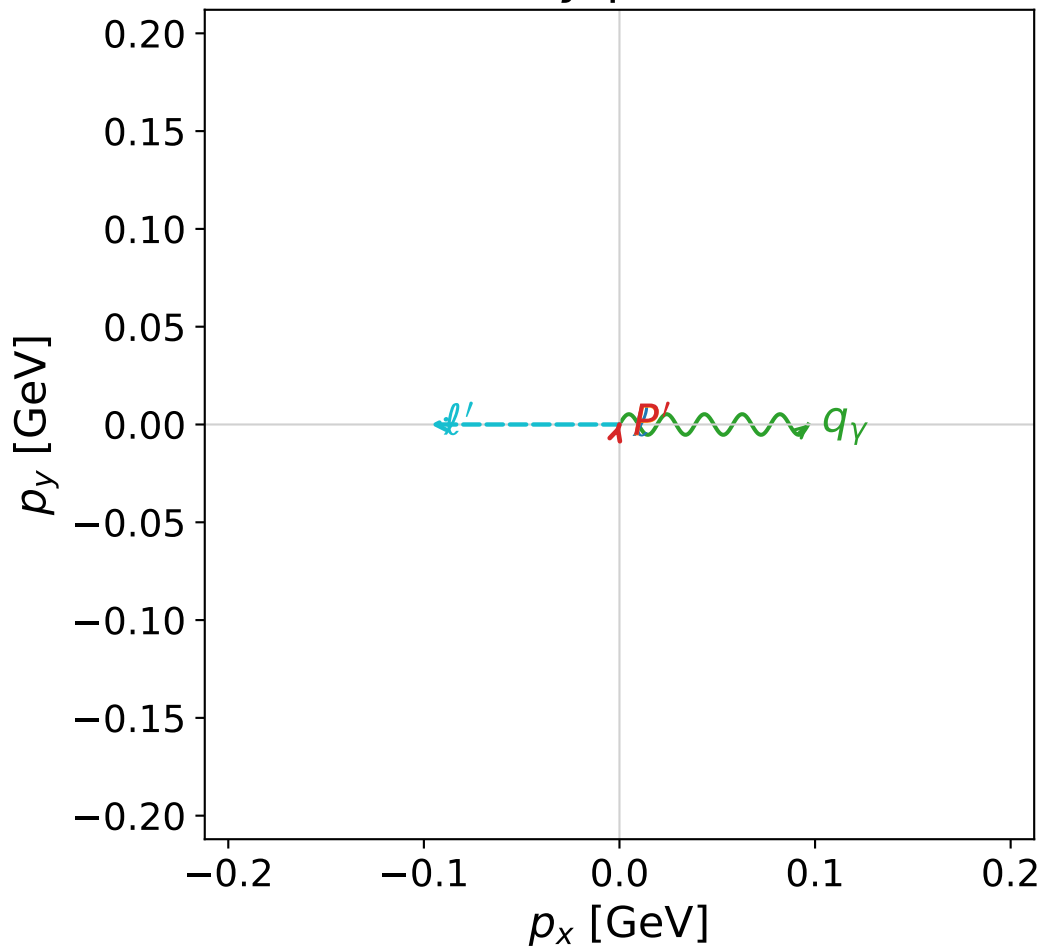


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



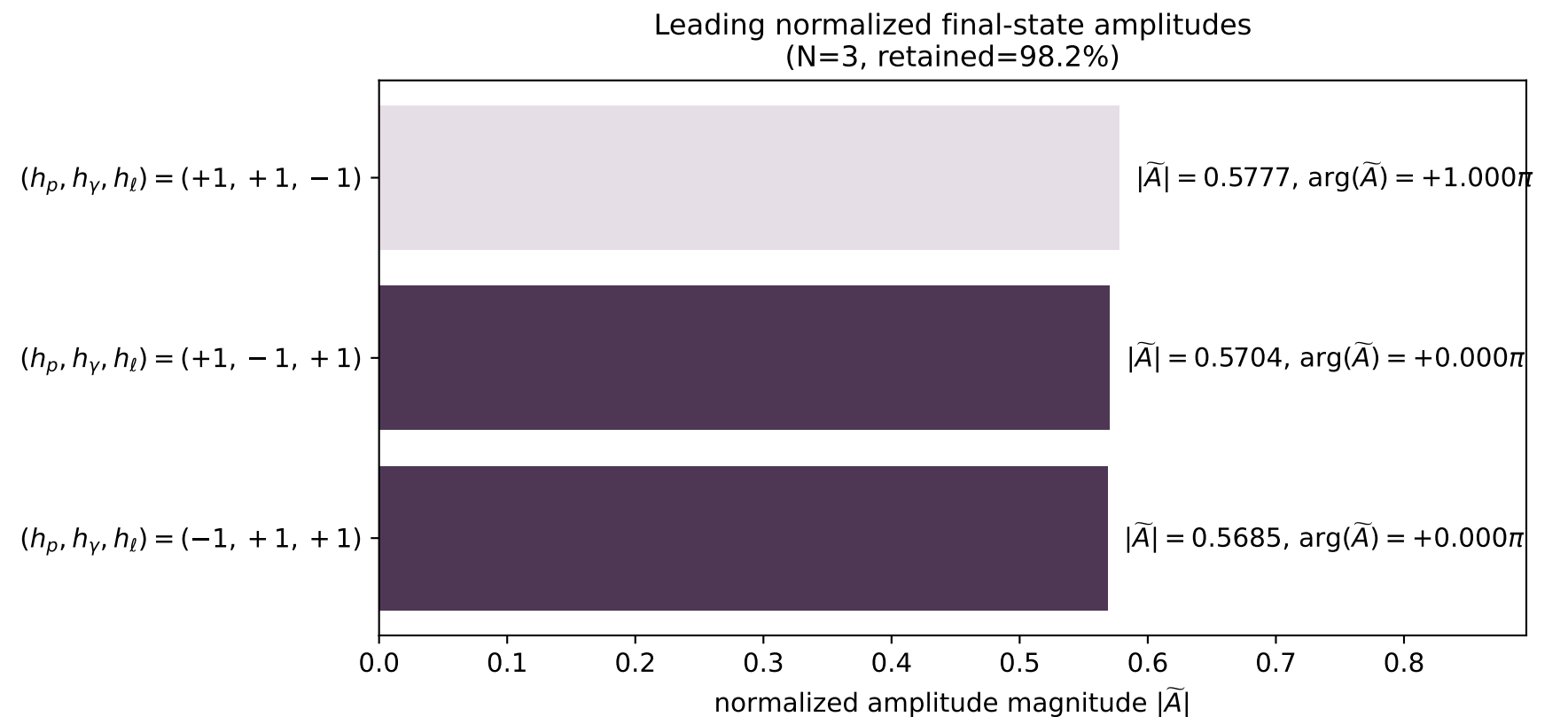
x--y plane



dw_mixing_angles_polarization_05_local_minimum_776
 $D_W=0.00968106$
 region: polarization_cluster_05_local_minimum_776
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0776
 lepton mixing angle: $\alpha_e=2.3464431$ rad
 incoming lepton: $-0.700178|+\rangle +0.713968|-\rangle$
 proton mixing angle: $\alpha_p=2.9353127$ rad
 incoming proton: $-0.9788|+\rangle +0.20482|-\rangle$

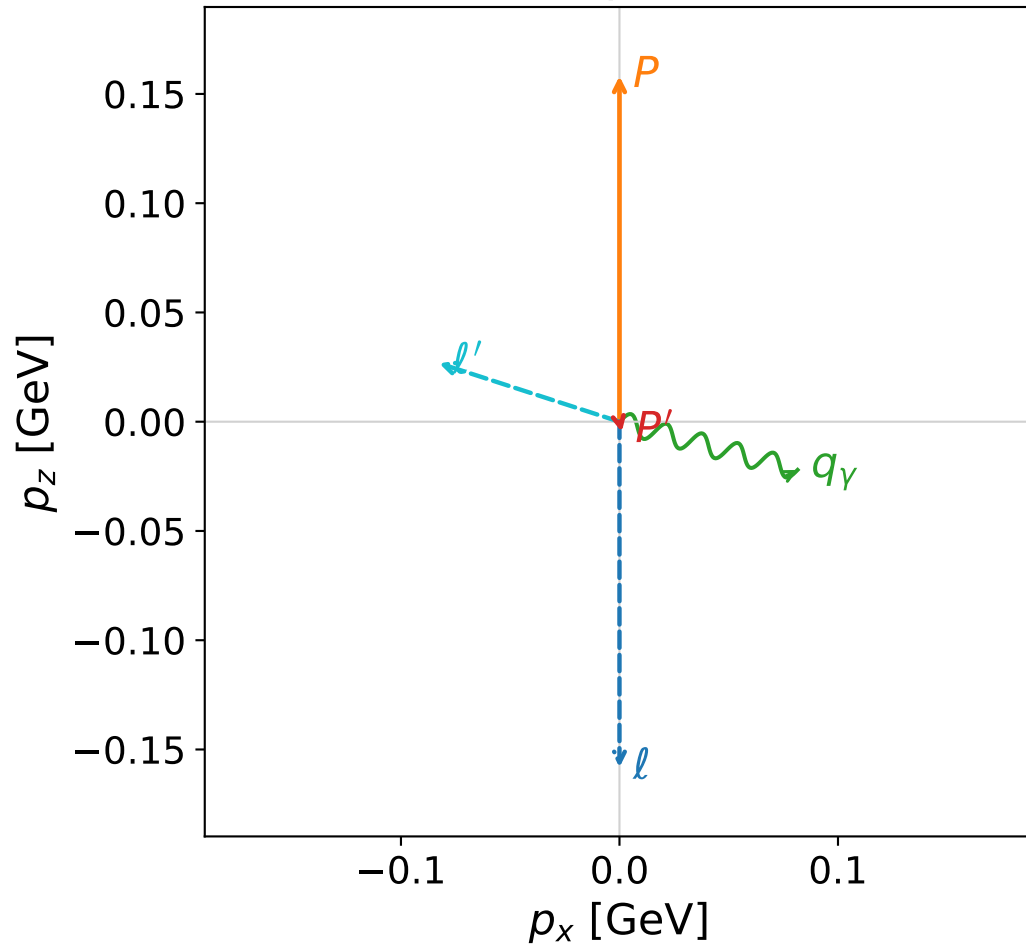
$s=1.27959$, $\sqrt{s}=1.13119$, $|\vec{P}|=0.176693$, $|\vec{P}'|=0.000242767$
 $\theta_{p'}=8.30818e-07$, $\theta_\gamma=1.53179$, $\phi_{p'}=1.27106$, $\phi_\gamma=6.28317$
 $E_\gamma=0.09659$, $\theta_{\ell'}=1.61231$, $\phi_{\ell'}=3.14157$
 $Q^2=0.0327203$, $x_B=0.152955$, $t=-0.0308627$, $W^2=1.06104$, $y=0.535139$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1767$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1767$
 P $E=0.9545$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1767$
 ℓ' $E=0.0966$ $p_x=-0.0965$ $p_y=0.0000$ $p_z=-0.0040$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.0002$
 q_γ $E=0.0966$ $p_x=0.0965$ $p_y=-0.0000$ $p_z=0.0038$

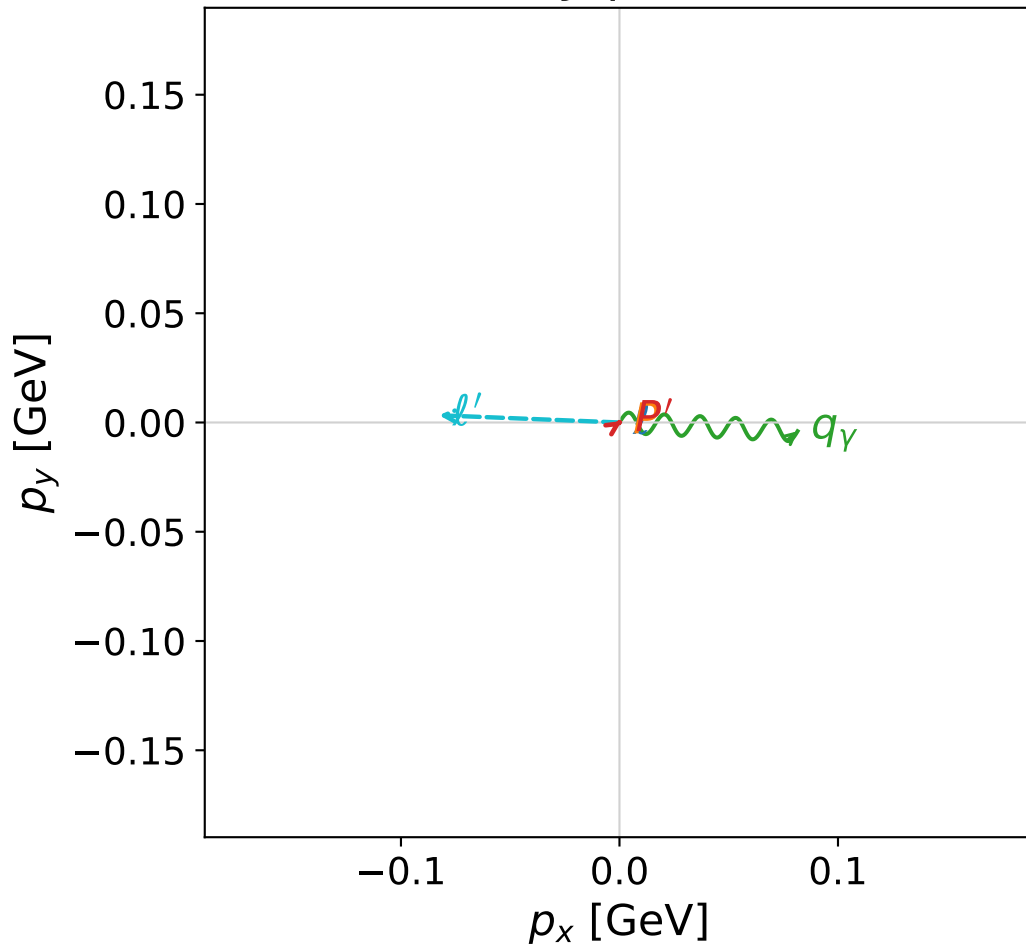


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_05_local_minimum_793

D_W=0.0102042

region: polarization_cluster_05_local_minimum_793

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0793

lepton mixing angle: $\alpha_e=2.3866647$ rad

incoming lepton: $-0.728321|+\rangle + 0.685236|-\rangle$

proton mixing angle: $\alpha_p=2.9399128$ rad

incoming proton: $-0.979731|+\rangle + 0.200315|-\rangle$

$s=1.23076$, $\sqrt{s}=1.1094$, $|\vec{P}|=0.158156$, $|\vec{P}'|=0.00484159$

$\theta_{p'}=2.88114$, $\theta_Y=1.83482$, $\phi_{p'}=0.579384$, $\phi_Y=6.23421$

$E_Y=0.0845404$, $\theta_{\ell'}=1.25779$, $\phi_{\ell'}=3.10149$

$Q^2=0.0359278$, $x_B=0.185045$, $t=-0.0263416$, $W^2=1.03807$, $y=0.553284$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1582$

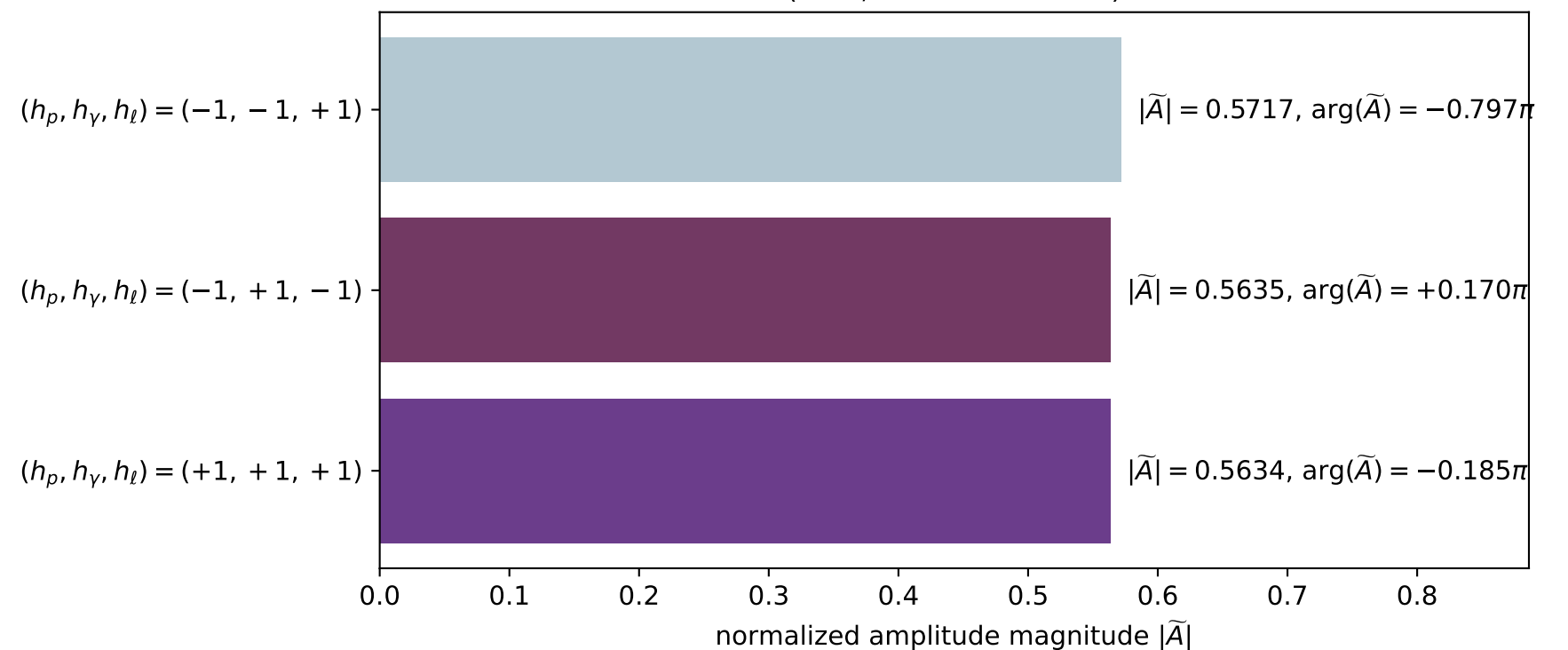
P $E= 0.9512$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1582$

ℓ' $E= 0.0868$ $p_x= -0.0826$ $p_y= 0.0033$ $p_z= 0.0267$

P' $E= 0.9380$ $p_x= 0.0010$ $p_y= 0.0007$ $p_z= -0.0047$

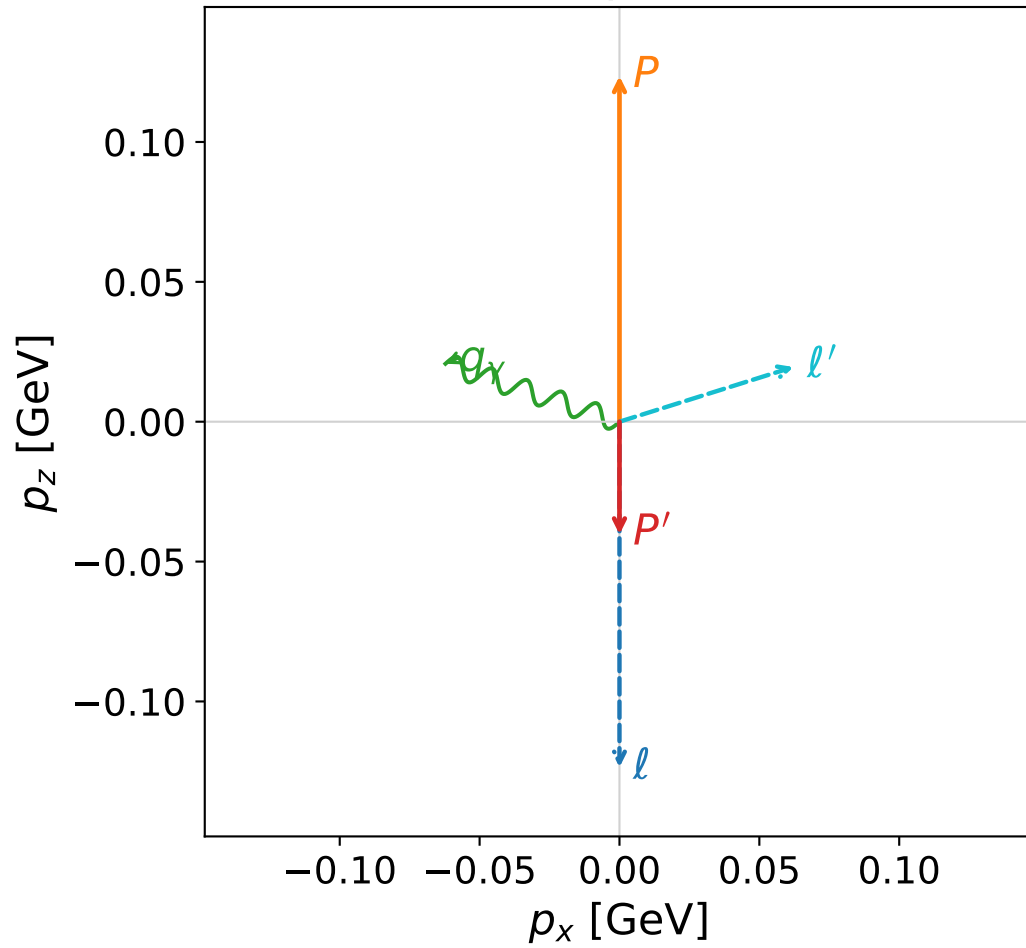
q_Y $E= 0.0845$ $p_x= 0.0815$ $p_y= -0.0040$ $p_z= -0.0221$

Leading normalized final-state amplitudes
(N=3, retained=96.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

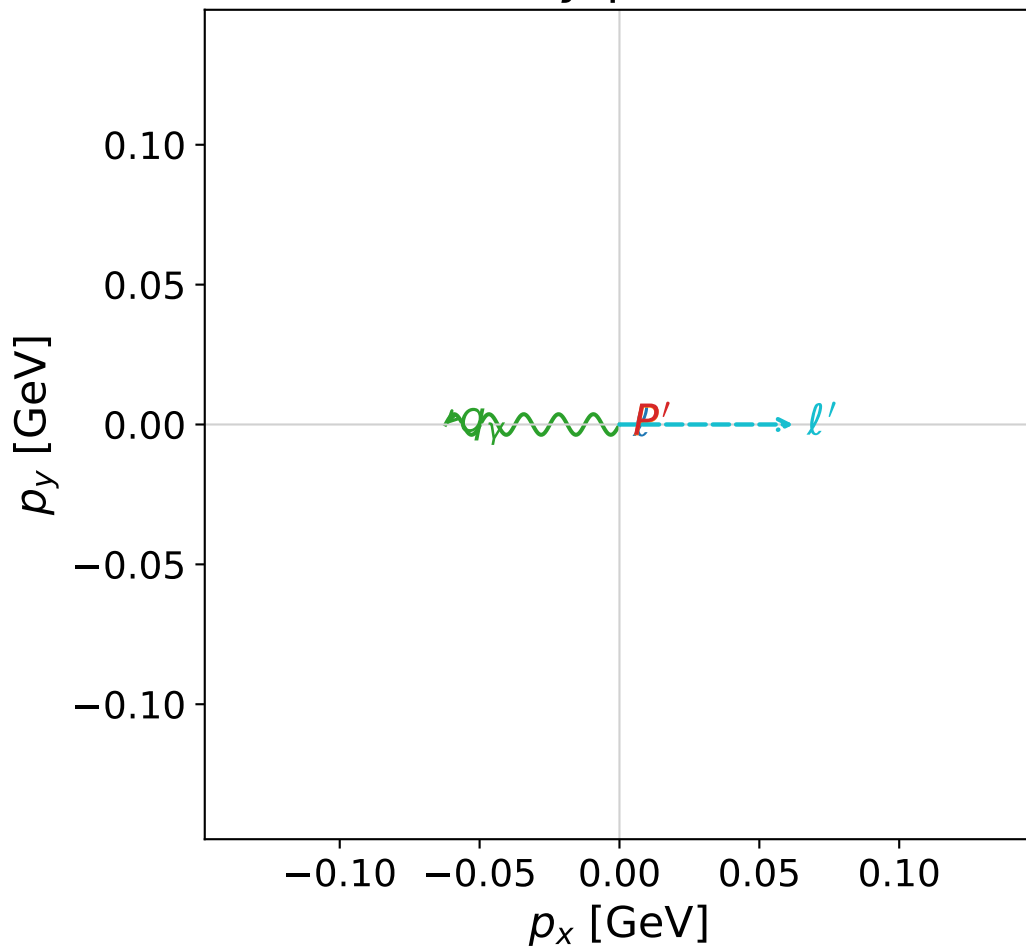


dw_mixing_angles_polarization_05_local_minimum_794
 $D_W=0.0102103$
 region: polarization_cluster_05_local_minimum_794
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0794
 lepton mixing angle: $\alpha_e=2.372709$ rad
 incoming lepton: $-0.718687|+\rangle + 0.695333|-\rangle$
 proton mixing angle: $\alpha_p=0.17283578$ rad
 incoming proton: $0.985101|+\rangle + 0.171977|-\rangle$

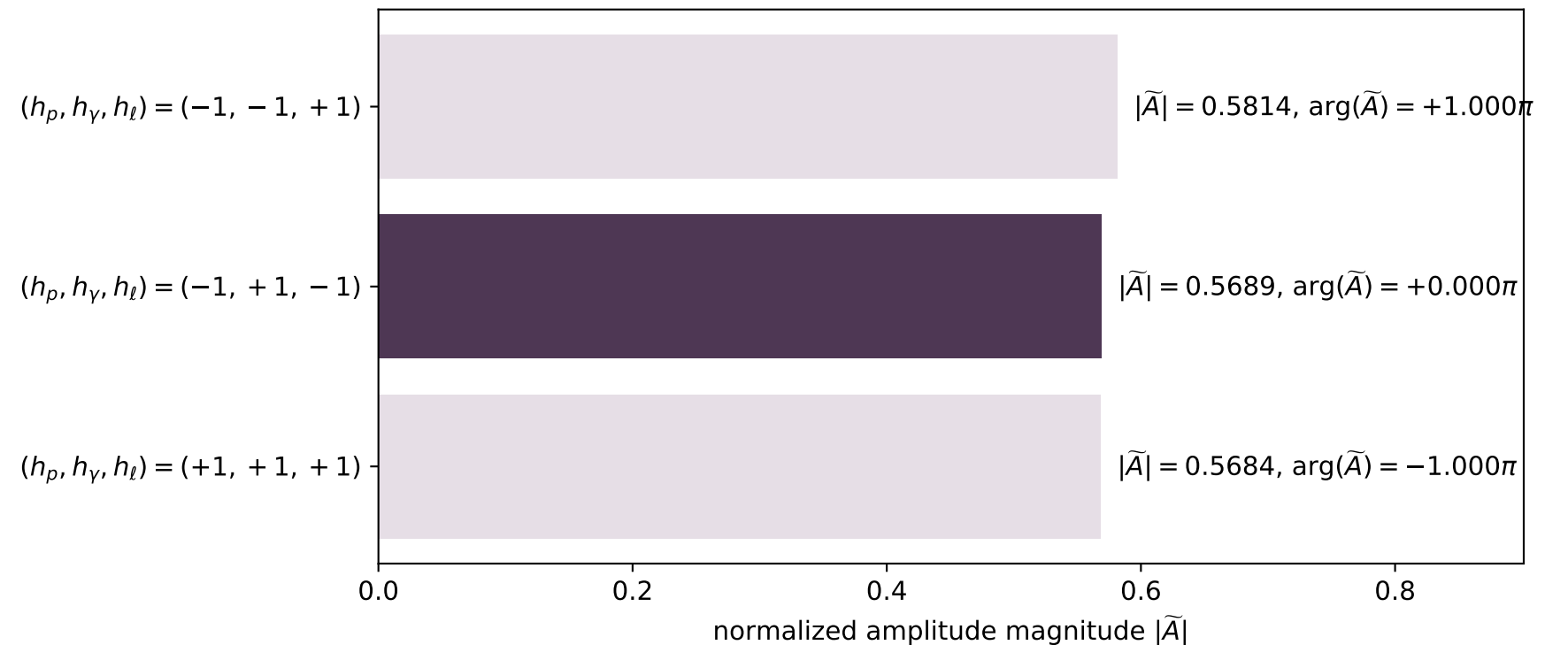
$s=1.14414$, $\sqrt{s}=1.06964$, $|\vec{P}|=0.123543$, $|\vec{P}'|=0.0402749$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.24932$, $\phi_{p'}=2.93467$, $\phi_\gamma=3.14162$
 $E_\gamma=0.0655678$, $\theta_{\ell'}=1.26619$, $\phi_{\ell'}=2.75644e-05$
 $Q^2=0.0209452$, $x_B=0.143724$, $t=-0.0267839$, $W^2=1.00463$, $y=0.551399$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1235$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1235$
 P $E= 0.9461$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1235$
 ℓ' $E= 0.0652$ $p_x= 0.0622$ $p_y= 0.0000$ $p_z= 0.0196$
 P' $E= 0.9389$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0403$
 q_γ $E= 0.0656$ $p_x= -0.0622$ $p_y= -0.0000$ $p_z= 0.0207$

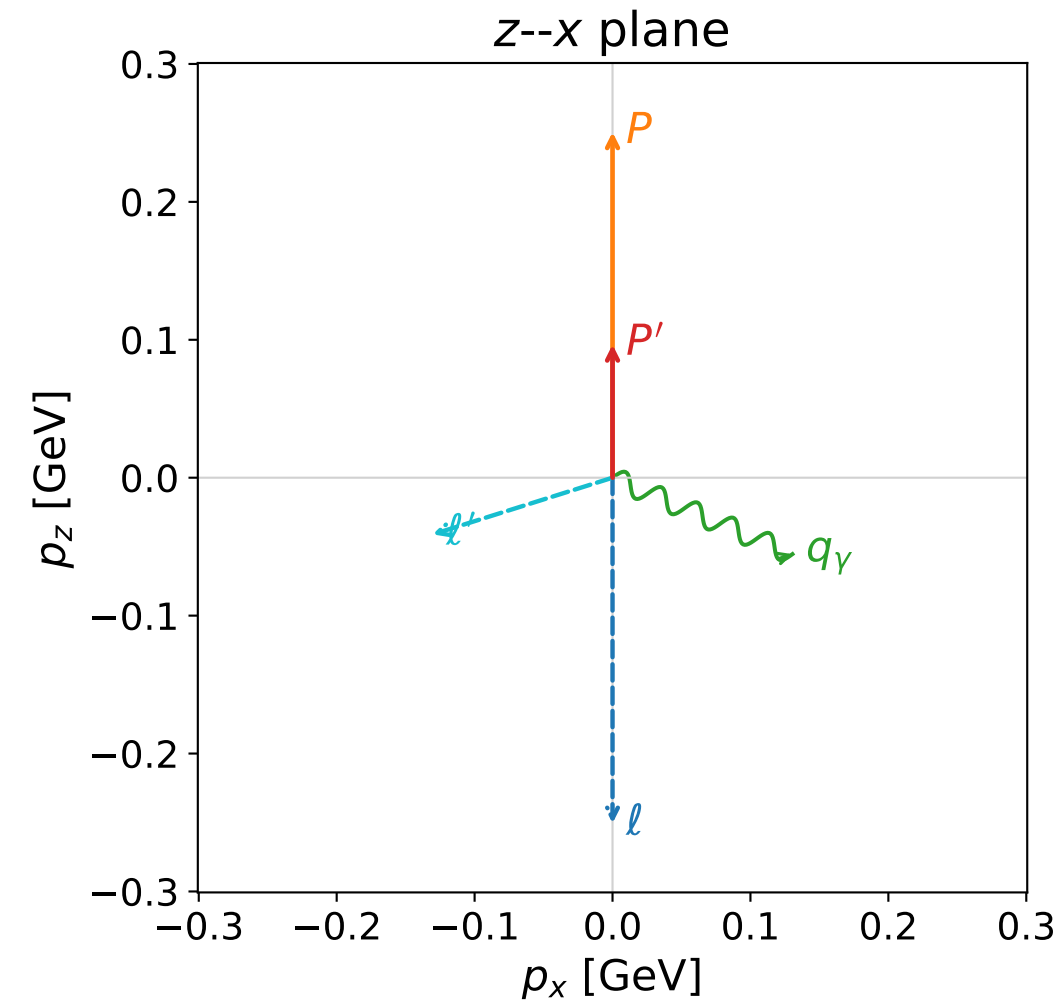
x--y plane



Leading normalized final-state amplitudes
 (N=3, retained=98.5%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

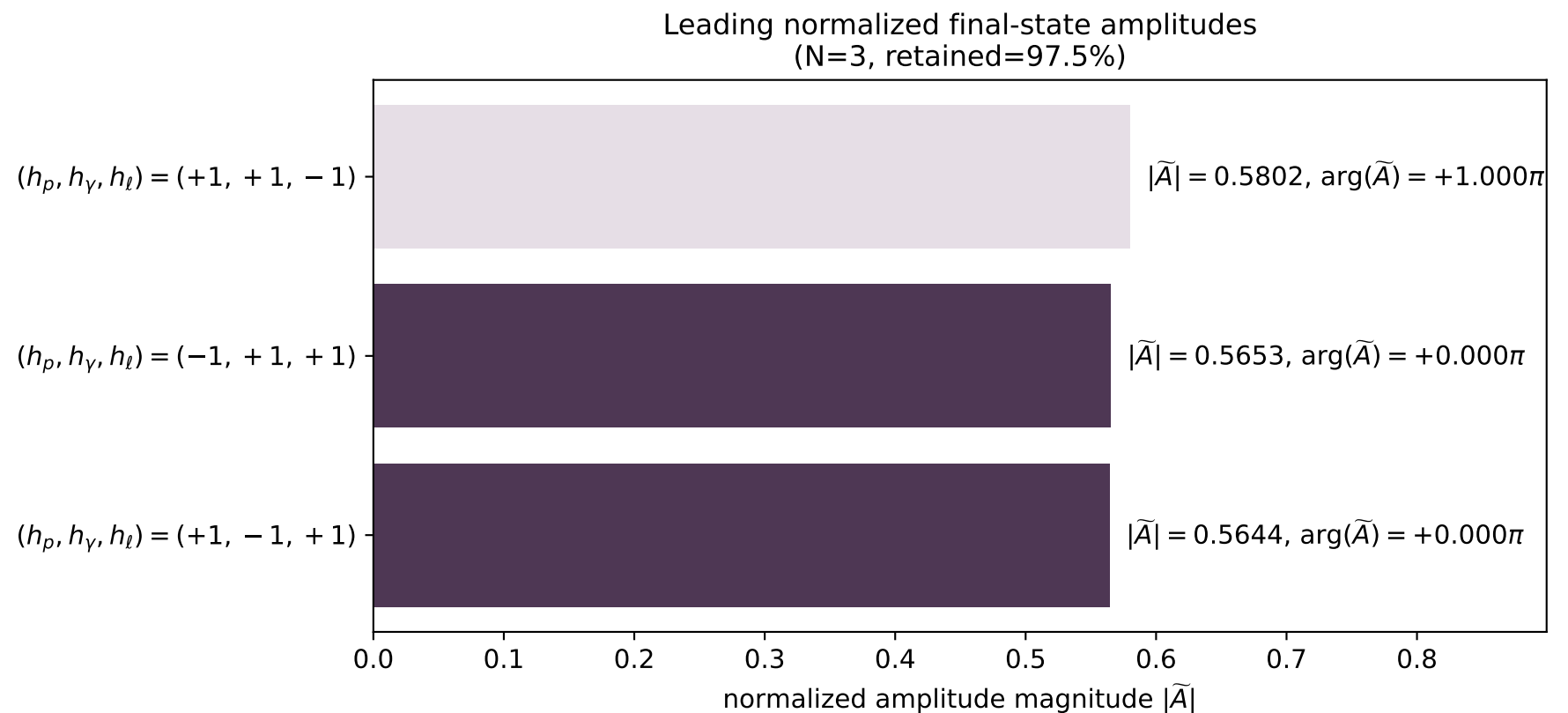
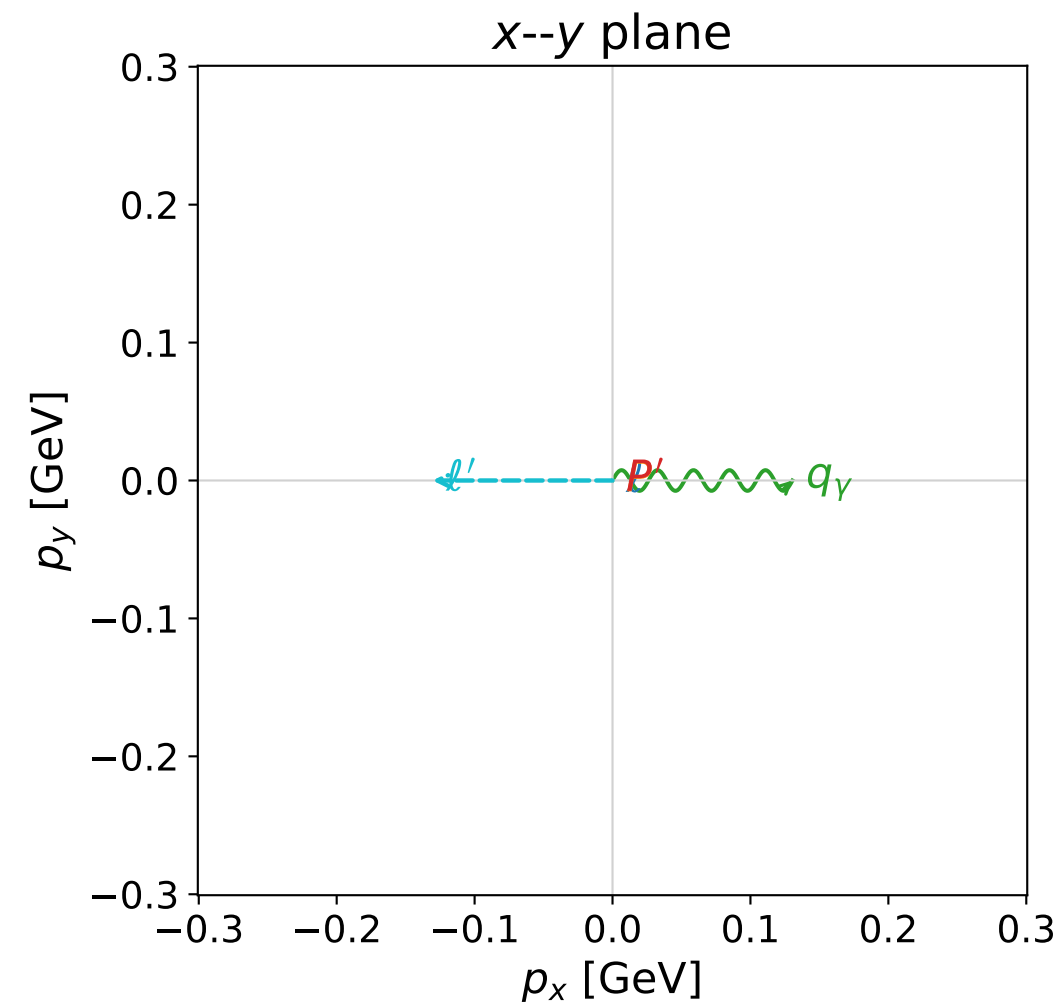


dw_mixing_angles_polarization_05_local_minimum_921
 $D_W=0.0145517$
 region: polarization_cluster_05_local_minimum_921
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0921
 lepton mixing angle: $\alpha_e=2.3318716$ rad
 incoming lepton: $-0.6897|+\rangle +0.724095|-\rangle$
 proton mixing angle: $\alpha_p=3.0185833$ rad
 incoming proton: $-0.992444|+\rangle +0.122699|-\rangle$

$s=1.49199$, $\sqrt{s}=1.22147$, $|\vec{P}|=0.250577$, $|\vec{P}'|=0.0965147$
 $\theta_{p'}=0$, $\theta_\gamma=1.97239$, $\phi_{p'}=4.7718$, $\phi_\gamma=6.28317$
 $E_\gamma=0.141734$, $\theta_{\ell'}=1.87609$, $\phi_{\ell'}=3.14157$
 $Q^2=0.0479455$, $x_B=0.1471$, $t=-0.0229545$, $W^2=1.15784$, $y=0.532451$

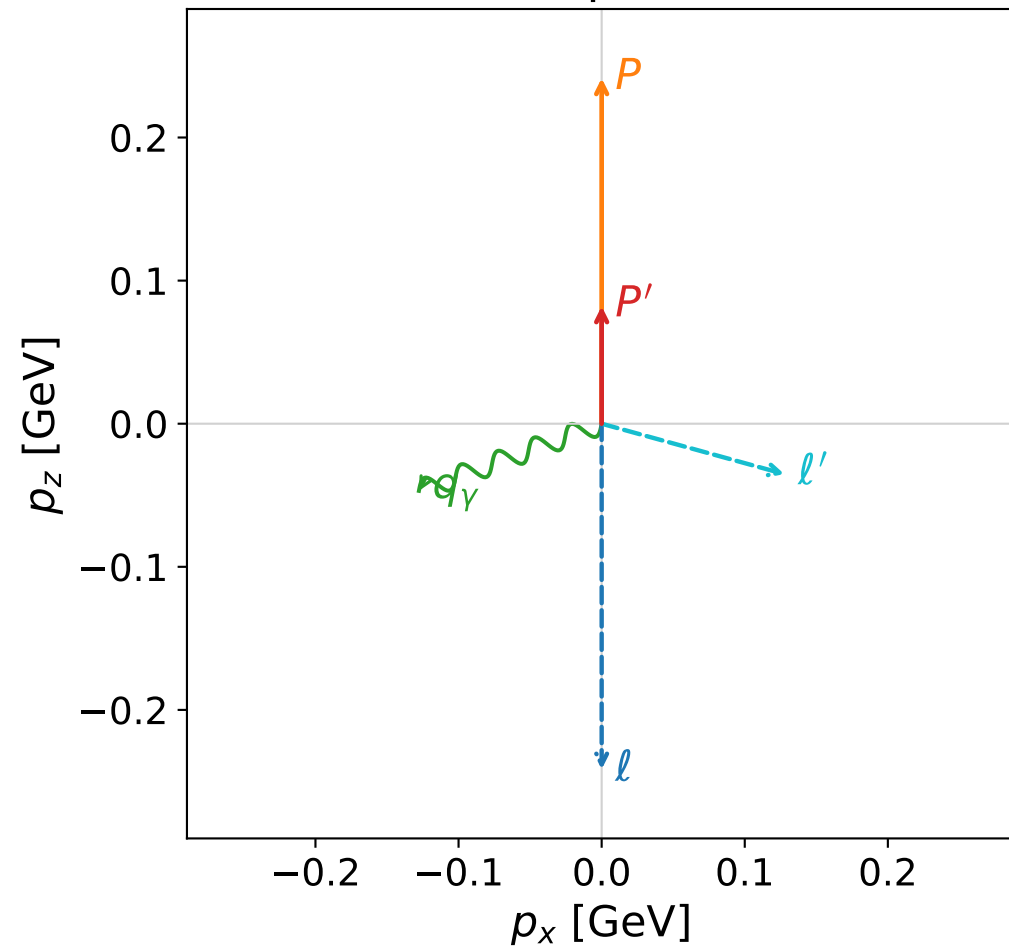
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.2506	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.2506
P	$E=$	0.9709	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.2506
ℓ'	$E=$	0.1368	$p_x=$	-0.1305	$p_y=$	0.0000	$p_z=$	-0.0411
P'	$E=$	0.9430	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	0.0965
q_γ	$E=$	0.1417	$p_x=$	0.1305	$p_y=$	-0.0000	$p_z=$	-0.0554

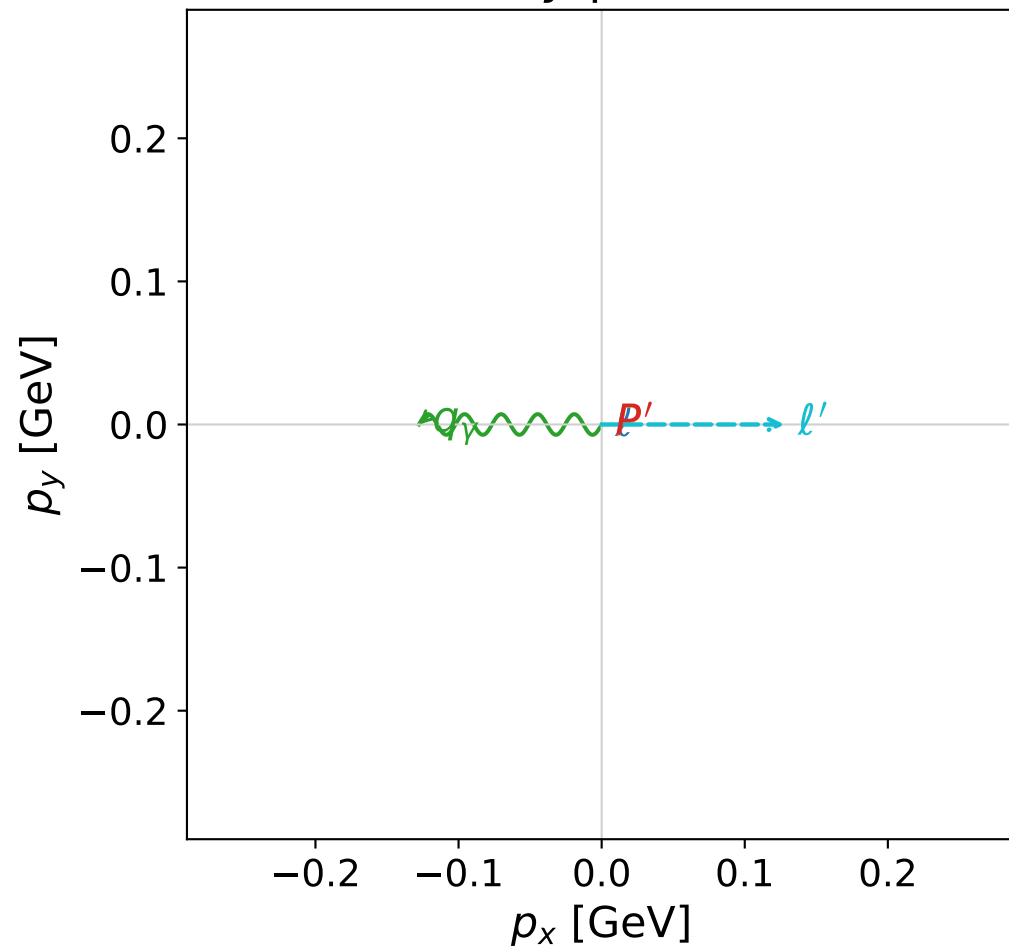


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

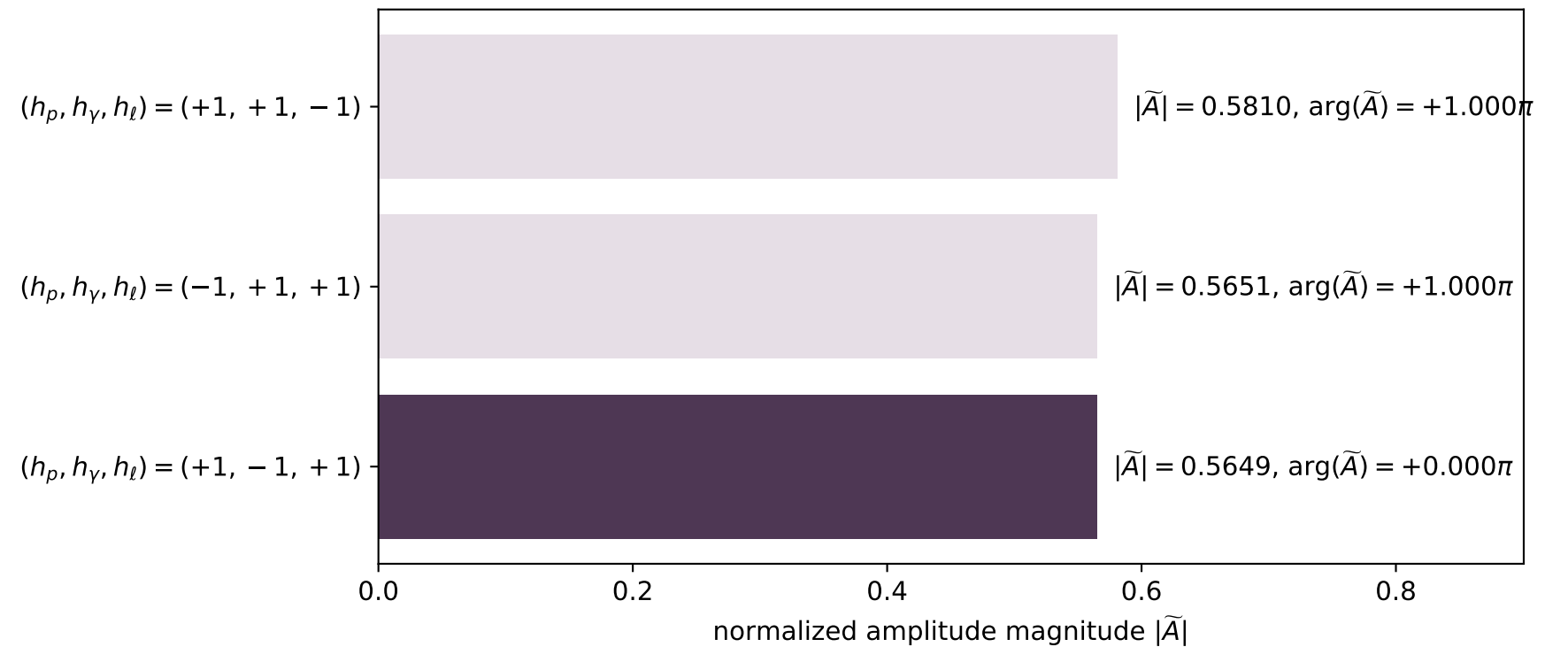


dw_mixing_angles_polarization_05_local_minimum_923
 $D_W=0.0146924$
 region: polarization_cluster_05_local_minimum_923
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0923
 lepton mixing angle: $\alpha_e=2.324786$ rad
 incoming lepton: $-0.684552|+\rangle +0.728964|-\rangle$
 proton mixing angle: $\alpha_p=0.13355488$ rad
 incoming proton: $0.991095|+\rangle +0.133158|-\rangle$

$s=1.46432$, $\sqrt{s}=1.21009$, $|\vec{P}|=0.241502$, $|\vec{P}'|=0.0821138$
 $\theta_{P'}=0$, $\theta_\gamma=1.92292$, $\phi_{P'}=3.23812$, $\phi_\gamma=3.1415$
 $E_\gamma=0.136046$, $\theta_{\ell'}=1.83971$, $\phi_{\ell'}=6.28309$
 $Q^2=0.0469803$, $x_B=0.151118$, $t=-0.0246754$, $W^2=1.14375$, $y=0.531898$

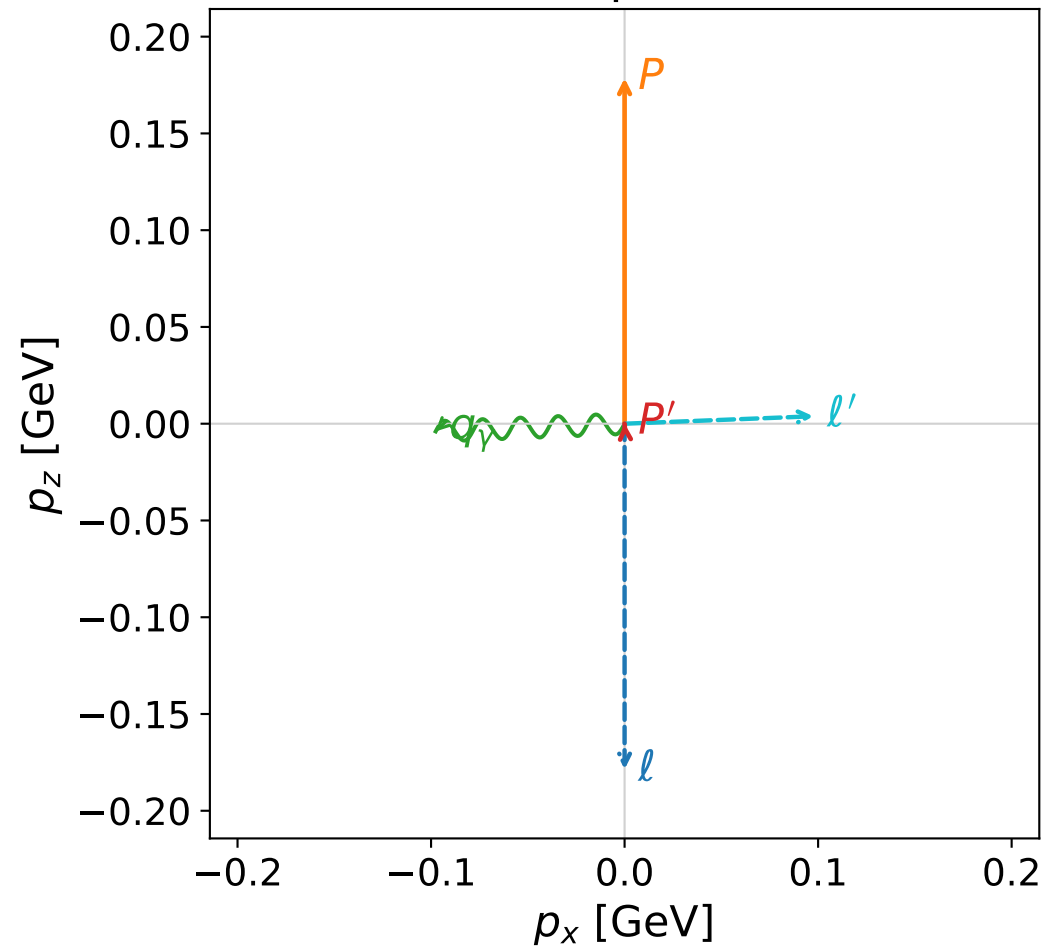
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.2415$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2415$
 P $E=0.9686$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2415$
 ℓ' $E=0.1325$ $p_x=0.1277$ $p_y=-0.0000$ $p_z=-0.0352$
 P' $E=0.9416$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0821$
 q_γ $E=0.1360$ $p_x=-0.1277$ $p_y=0.0000$ $p_z=-0.0469$

Leading normalized final-state amplitudes
 (N=3, retained=97.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



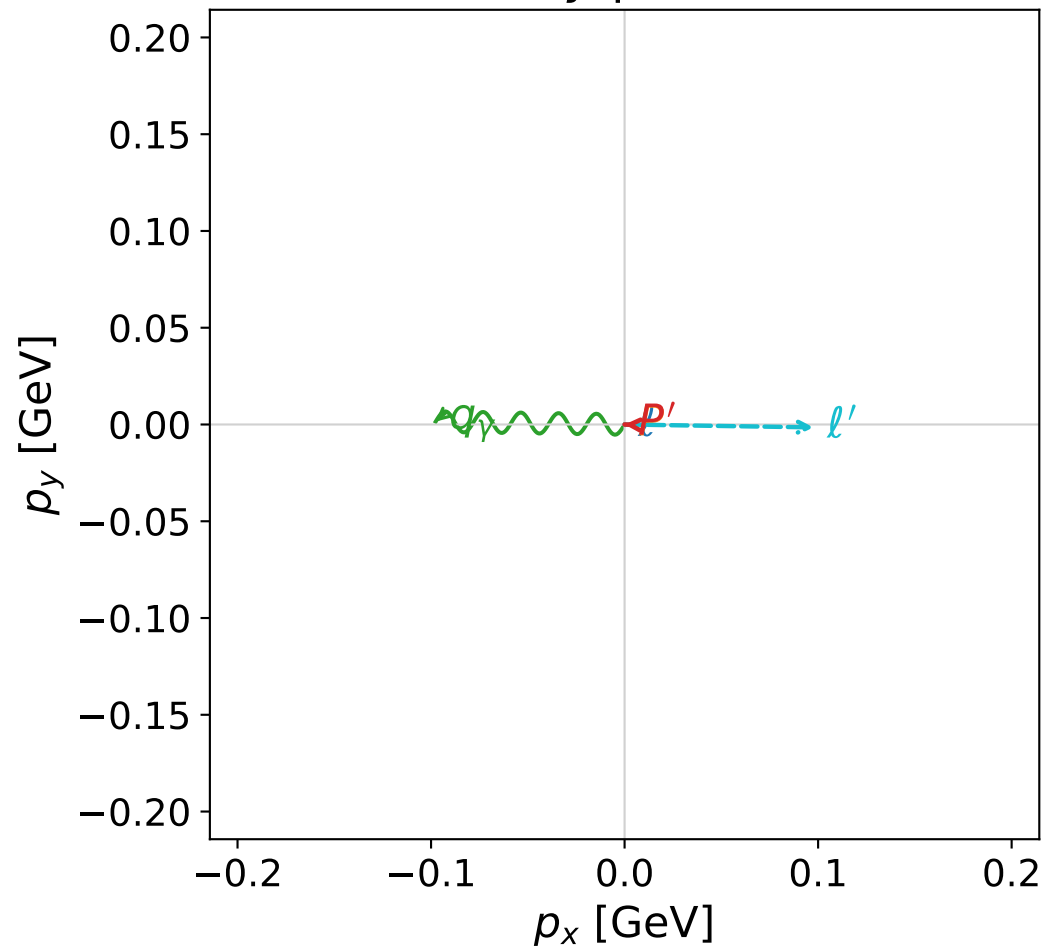
dw_mixing_angles_polarization_05_local_minimum_1091
 $D_W=0.0262184$
 region: polarization_cluster_05_local_minimum_1091
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_1091
 lepton mixing angle: $\alpha_e=2.3526069$ rad
 incoming lepton: $-0.704565|+\rangle +0.709639|-\rangle$
 proton mixing angle: $\alpha_p=0.0056745158$ rad
 incoming proton: $0.999984|+\rangle +0.00567449|-\rangle$

$s=1.28465$, $\sqrt{s}=1.13342$, $|\vec{P}|=0.178576$, $|\vec{P}'|=0.000129391$
 $\theta_{p'}=0.0541711$, $\theta_\gamma=1.61158$, $\phi_{p'}=3.10435$, $\phi_\gamma=3.12631$
 $E_\gamma=0.0977106$, $\theta_{\ell'}=1.53134$, $\phi_{\ell'}=6.2679$
 $Q^2=0.036275$, $x_B=0.165202$, $t=-0.0315595$, $W^2=1.06315$, $y=0.542431$

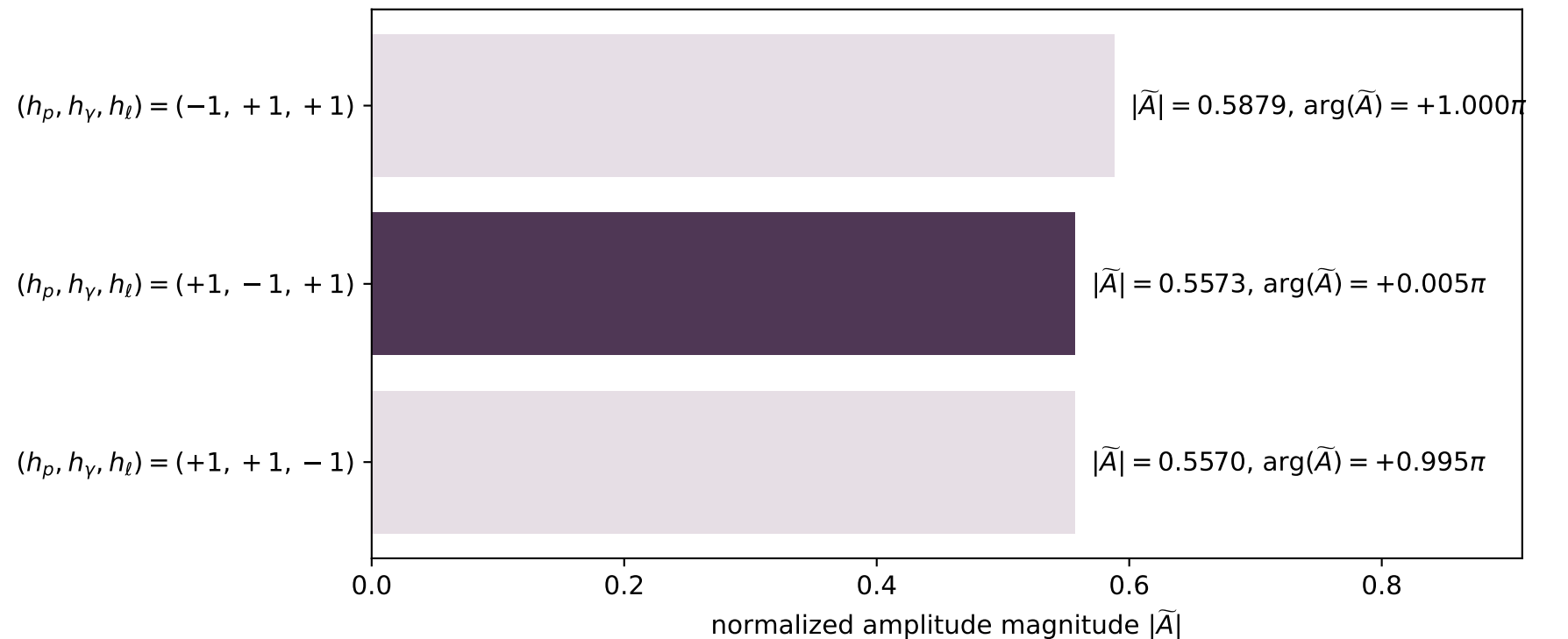
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1786	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1786
P	$E=$	0.9548	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1786
ℓ'	$E=$	0.0977	$p_x=$	0.0976	$p_y=$	-0.0015	$p_z=$	0.0039
P'	$E=$	0.9380	$p_x=$	-0.0000	$p_y=$	0.0000	$p_z=$	0.0001
q_γ	$E=$	0.0977	$p_x=$	-0.0976	$p_y=$	0.0015	$p_z=$	-0.0040

x--y plane

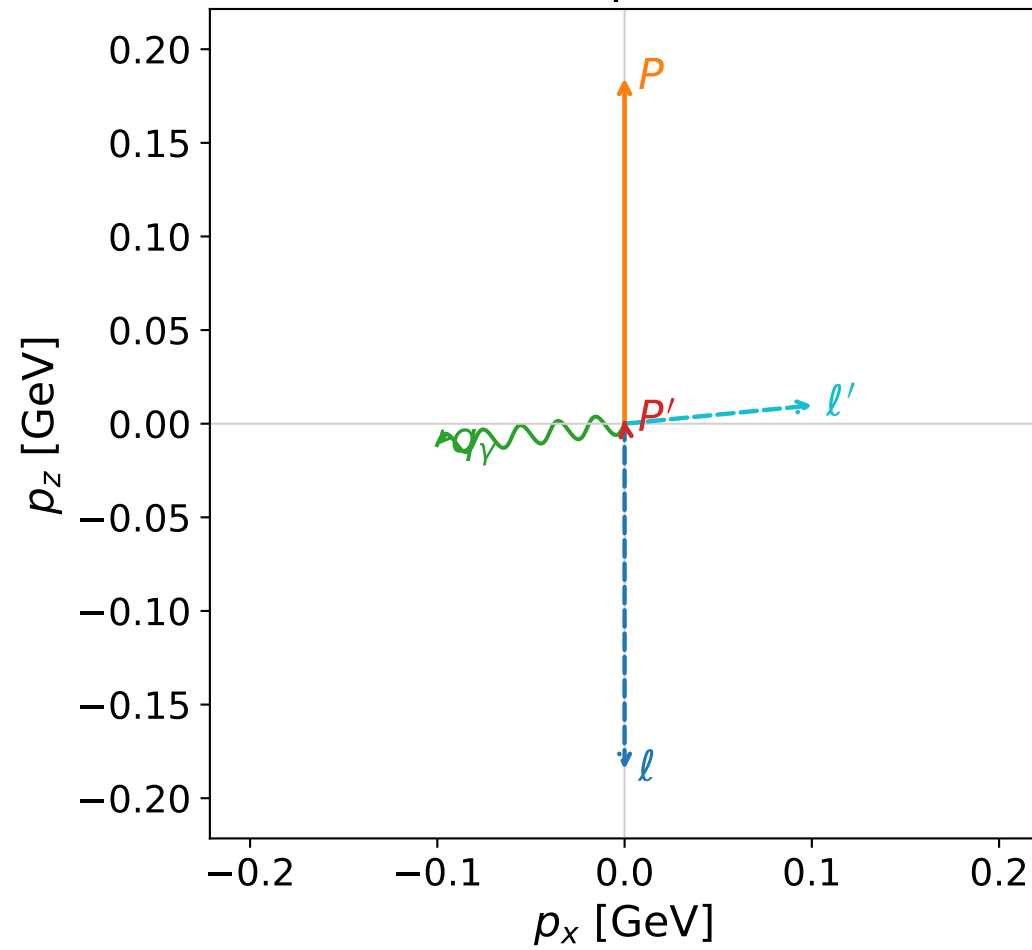


Leading normalized final-state amplitudes
(N=3, retained=96.6%)

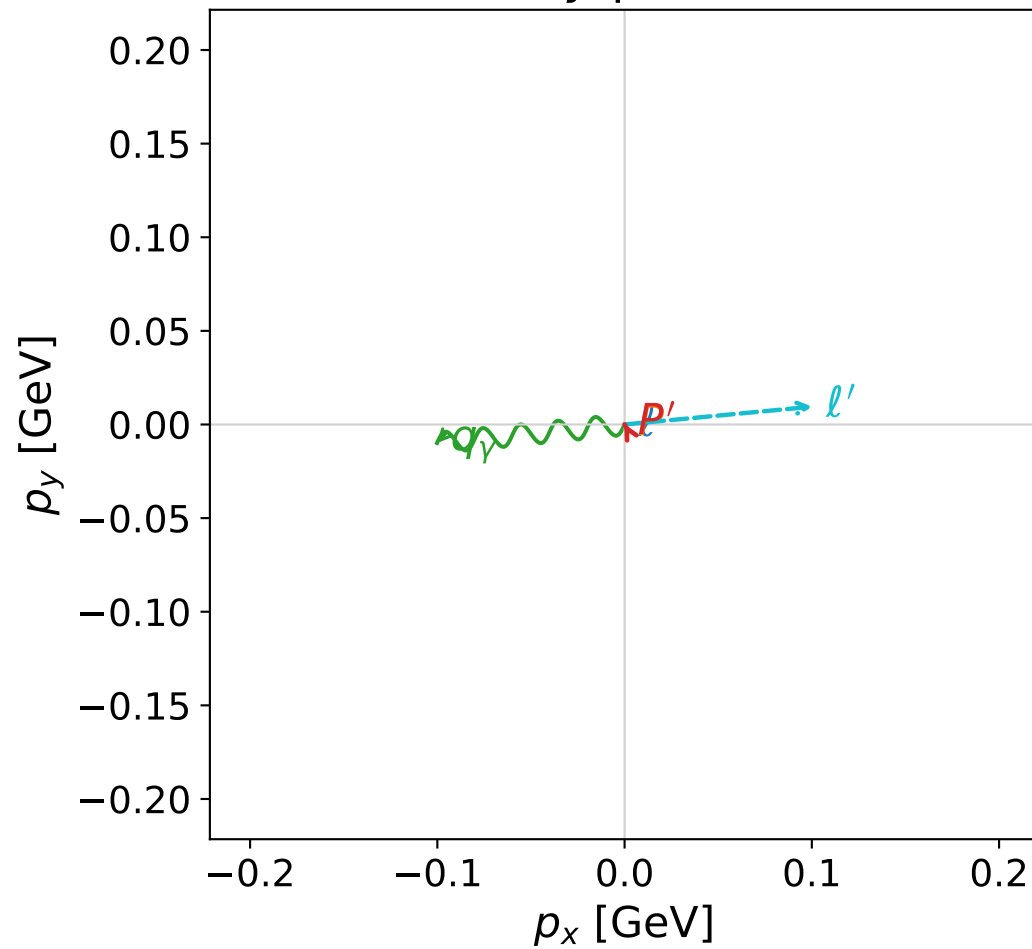


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_05_local_minimum_1123
 $D_W=0.031282$
 region: polarization_cluster_05_local_minimum_1123
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_1123
 lepton mixing angle: $\alpha_e=2.3460623$ rad
 incoming lepton: $-0.699906|+\rangle +0.714235|-\rangle$
 proton mixing angle: $\alpha_p=0.0032211657$ rad
 incoming proton: $0.999995|+\rangle +0.00322116|-\rangle$

$s=1.30086$, $\sqrt{s}=1.14055$, $|\vec{P}|=0.184565$, $|\vec{P}'|=0.0016487$
 $\theta_{p'}=0.239026$, $\theta_\gamma=1.68355$, $\phi_{p'}=2.06697$, $\phi_\gamma=3.23994$
 $E_\gamma=0.101283$, $\theta_{l'}=1.47393$, $\phi_{l'}=0.094778$
 $Q^2=0.040996$, $x_B=0.177465$, $t=-0.0331524$, $W^2=1.06986$, $y=0.548696$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1846$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1846$
 P $E=0.9560$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1846$
 l' $E=0.1013$ $p_x=0.1003$ $p_y=0.0095$ $p_z=0.0098$
 P' $E=0.9380$ $p_x=-0.0002$ $p_y=0.0003$ $p_z=0.0016$
 q_γ $E=0.1013$ $p_x=-0.1002$ $p_y=-0.0099$ $p_z=-0.0114$

Leading normalized final-state amplitudes
 (N=3, retained=95.8%)

